
TSFMI: A Baseline-Controlled Evaluation Protocol for Time-Series Foundation Model Representations

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Abstract

1 We introduce a baseline-controlled evaluation protocol for time-series foundation
2 model (TSFM) representation probing, applied to seven pre-trained model families
3 and six temporal properties under a canonical 60/20/20 train/val/test split with
4 matched sklearn estimators, 5 seeds, and bootstrap 95% CIs. Our headline empirical
5 finding is a *performance inversion*: on anomaly probing – the only property with
6 meaningful model variance (0.50–0.75 across families) – hand-crafted statistical
7 features reach 0.858 and outperform every pre-trained TSFM. The mechanism
8 is direct: anomaly classification on this task reduces to recovering kurtosis (a
9 single HC feature reaches 0.859 and a single max-magnitude feature 0.907 on
10 the same protocol), and six of seven TSFMs cannot regress kurtosis from their
11 own representations ($R^2 \leq 0.22$). The remaining four standard properties (trend,
12 frequency, stationarity, change point) saturate at ≥ 0.999 for both non-model
13 baselines and TSFMs and function as ceiling tests for the protocol rather than
14 discriminators between models; the empirical signal of the paper is on anomaly
15 and seasonality. Seasonality is artifact-sensitive: a negative PCA-probe R^2 inverts
16 to > 0.999 under full-dimensional Ridge, indicating an evaluation-pipeline artifact
17 rather than a representation deficit. Bounded diagnostics: (i) capacity-sweep
18 MLP probes (64–1024 units) plus Random Forest on TSFM representations do
19 not exceed HC on anomaly; (ii) a 10-feature `tsfresh-minimal` baseline reaches
20 0.899 and a 1024-kernel ROCKET-style baseline 0.813 (anomaly) / 0.960 (anomaly-
21 hard); (iii) a 5×5 (data-seed $\times$ split-seed) bootstrap holds the inversion at HC
22 = 0.871 [0.862, 0.880] over 25 cells; (iv) a literal reproduction of Wiliński et al.
23 2025 shows HC at or above their MOMENT Fisher-LDR scores. The takeaway is
24 methodological: TSFM representation claims require explicit non-model baselines,
25 artifact checks, and bounded intervention diagnostics. We release the protocol,
26 scripts, and 170 automated tests as anonymized supplementary material; reviewers
27 can reproduce one cell end-to-end via `make smoke`.

28 1 Introduction

29 Time Series Foundation Models (TSFMs) achieve strong downstream performance on forecasting,
30 classification, and anomaly detection. A rapidly growing literature probes their frozen representations
31 to ask what temporal concepts they “encode”. Such probing claims, however, are often evaluated
32 without the explicit control-task discipline that has been standard in NLP probing since Hewitt and
33 Liang [1]: high probe accuracy alone does not demonstrate model-specific knowledge if the same
34 property is already linearly recoverable from raw input signals or a handful of hand-crafted statistics.
35 We make this gap concrete with one empirical anchor result and a matched evaluation protocol built
36 around it.

Headline result. Under a canonical 60/20/20 train/val/test protocol with matched sklearn estimators, 5 seeds, and bootstrap 95% CIs applied identically to non-model baselines and to probes of seven pre-trained TSFM families, we find a performance inversion on anomaly detection: an 8-dim hand-crafted feature vector reaches 0.858 test accuracy and *outperforms every one of seven pre-trained TSFMs* (best model 0.753). Anomaly is also the only property on which model probes vary meaningfully; the other four standard synthetic properties (trend, frequency, stationarity, change point) are already saturated (≥ 0.999) by non-model baselines under the current task definitions. Seasonality additionally exhibits an evaluation-pipeline artifact: negative PCA-probe R^2 becomes > 0.999 once a matched Ridge replaces the original probe setup. Taken together, these results show that probe accuracy on most standard TSFM probing tasks cannot, on its own, be interpreted as evidence of model-specific temporal encoding.

We introduce TSFMI, a benchmark protocol for evaluating temporal concept accessibility in TSFM representations under explicit baseline controls. The framework supports cross-family comparison through a shared extraction interface (4 methods; existing wrappers range 121–299 lines, median 173) and a common probing, intervention, and similarity pipeline. TSFMI standardizes linear probing, LEACE concept erasure, LDA steering, structural probes, and CKA into one reusable protocol spanning native TSFMs (MOMENT, Chronos-Bolt, Timer, TimesFM, Moirai), a Transformer-TS model (PatchTST), and an LLM-adapted model (GPT4TS). Our goal is not to claim exact hidden-state equivalence across these families, but to provide a common protocol for comparing standardized extracted views under matched controls.

We evaluate six temporal properties across synthetic datasets with exact ground truth and five real-world benchmarks, totaling over 5,000 layer-level probe training runs (see Appendix A.1 for the auto-generated manifest), and report these as reference benchmark results. Our contribution is not that TSFMs fail, but that stronger controls are necessary before representation-level claims can be interpreted as model-specific. The paper makes the following contributions:

1. A **baseline-controlled benchmark protocol** covering 7 pre-trained architectures, 5 random baselines, and explicit non-model controls across 6 temporal properties with synthetic, hard-variant, and real-world datasets.¹
2. **Reference results under this protocol** showing that several commonly probed properties saturate strong non-model baselines, that selected hard/diverse variants reduce ceiling saturation, and that synthetic-to-real transfer remains weak.
3. An **intervention-based diagnostic suite** (LEACE erasure, LDA-subspace projection, random-subspace controls, entanglement analysis, nonlinear recovery) as secondary interpretive tools, with conclusions explicitly bounded by linearity assumptions and by the heterogeneous extraction choices required across model families.
4. **Open-source extensible infrastructure:** synthetic generators, extraction pipelines, evaluation scripts, and 170 automated tests for extension to new TSFMs.

2 Related Work

2.1 Time Series Foundation Models

MOMENT [2] introduced an encoder-only masked time series model pre-trained on the Time Series Pile. Chronos [3] tokenizes time series values into discrete bins using a T5-based architecture. TimesFM [4] adopts a decoder-only architecture with input patching, while Moirai [5] introduces a universal forecasting transformer. PatchTST [6] established channel-independent patching with self-supervised pre-training, and iTransformer [7] inverts the attention axis to attend over variates.

2.2 LLM Repurposing for Time Series

GPT4TS [8] demonstrated that a frozen GPT-2 backbone with only input/output layers fine-tuned achieves competitive forecasting performance. Time-LLM [9] aligns time series patches with text

¹Confirmatory: 42 canonical (model, property) cells + 6 baseline-control cells; secondary: 690 LEACE erasures, 60 CKA matrices, 33 intervention runs, 7 hard-variant comparisons. Auto-generated by `scripts/count_experiments.py`; see Appendix A.1.

84 prototypes. Tan et al. [10] challenged whether LLMs are “actually useful” for time series, while
 85 Schumacher et al. [11] showed that linear probes over LLM representations reach F1 of 0.61–0.67,
 86 far exceeding zero-shot prompting (0.15–0.26).

87 2.3 Probing and Representation Analysis

88 Belinkov [12] and Rogers et al. [13] survey probing in NLP, establishing the “BERTology” paradigm.
 89 Tenney et al. [14] show that BERT rediscovers the classical NLP pipeline across layers, while
 90 Conneau et al. [15] and Dalvi et al. [16] probe sentence and neuron-level representations. Hewitt and
 91 Liang [1] address probe capacity through control tasks. Voita and Titov [17] and Pimentel et al. [18]
 92 formalize information-theoretic alternatives, highlighting that high probe accuracy can reflect probe
 93 capacity rather than genuine encoding; our use of MLP control probes and LEACE erasure partially
 94 addresses this, though MDL-based probing remains a promising extension. Raghu et al. [19] and
 95 Morcos et al. [20] use CKA and canonical correlation to compare representations. Geiger et al. [21]
 96 formalize causal abstraction for neural networks, motivating our intervention-based validation.

97 For TSFMs, Wilinski et al. [22] analyze MOMENT with CKA and steering, Pandey et al. [23] study
 98 linear recoverability of temporal concepts, Kalnāre et al. [24] adapt mechanistic interpretability,
 99 and Oublal et al. [25] apply sparse autoencoders to time series explanations. Our contribution is
 100 protocol-level consolidation rather than a replacement of these prior mechanistic analyses. TSFMI
 101 complements them by providing a unified, reusable protocol that combines cross-family comparison,
 102 baseline controls, intervention validation, and hard-variant difficulty control under a single benchmark,
 103 as summarized below:

	TSFMI	Wiliński’25	Pandey’25	Kalnāre’25	Oublal’26
Model families	7	1	3	1	1
Temporal properties	6 + hard	3	4	—	—
Baseline controls	✓	—	—	—	—
Concept erasure	✓	—	—	—	—
Intervention validation	✓	✓	—	—	—
Hard variants	✓	—	—	—	—
Code released	✓	—	—	—	—

105 2.4 Concept Erasure and Representation Engineering

106 INLP [26] iteratively projects onto null spaces, and Elazar et al. [27] extend this with amnesic probing
 107 for behavioral explanation. LEACE [28] provides closed-form optimal erasure. Steering vectors
 108 [29, 30] control LLM behavior via activation addition and representation engineering. Aksu et al.
 109 [31] introduce GIFT-Eval, a comprehensive benchmark for evaluating TSFMs across 23 datasets. We
 110 extend this line of work with LEACE erasure and LDA steering applied to temporal concepts under a
 111 common protocol.

112 Existing time-series benchmark suites such as GIFT-Eval [31], Monash Archive [32], Darts [33], and
 113 anomaly detection benchmarks [34] focus on end-task comparison. Self-supervised TS representation
 114 methods (TS2Vec [35], TNC [36]) and classification surveys [37] evaluate learned representations
 115 but not through the lens of concept-level probing and intervention. Additional TSFMs such as
 116 Lag-Llama [38] and UniTS [39] further motivate cross-family evaluation. Widely used architectures
 117 such as TimesNet [40] prioritize model training/evaluation pipelines for downstream tasks. TSFMI
 118 complements these resources by benchmarking internal representations directly: it standardizes
 119 concept-level accessibility tests, intervention controls, entanglement diagnostics, and cross-model
 120 geometry analysis rather than only reporting endpoint task scores.

121 3 Benchmark Design

122 3.1 Problem Formulation

123 We consider a pre-trained time series model M with L layers. For input $X \in \mathbb{R}^{T \times C}$, we extract
 124 frozen representations $h^{(l)} \in \mathbb{R}^{N \times D}$ from each layer l , where N is the number of patches and D the
 125 hidden dimension. We probe these for temporal properties $P = \{p_1, \dots, p_k\}$.

Table 1: Extraction and reduction protocol by model family. Cross-family comparisons are valid only at the level of these standardized views, not raw hidden states. *Confirmatory benchmark result.*

Model	Internal Object	Aggregation	Dim. Reduction	Random Baseline?	Caveat
MOMENT	Encoder block outputs	None	PCA512	No ^a	PCA artifact on seasonality
Chronos-Bolt	Encoder block outputs	None	None	No ^a	Enc-dec bridge in CKA
PatchTST	Encoder layer outputs	None	None	Yes	Smallest model (1.5M)
GPT4TS	Transformer hidden states	None	PCA512	No ^a	LLM backbone frozen
Timer	Decoder layer outputs	Mean-pool	None	No ^b	Requires checkpoint init
TimesFM	Decoder layer outputs	Mean-pool	None	No ^b	50 layers, largest model
Moirai	Encoder layer outputs	Mean-pool	None	No ^b	Clusters w/ decoders in CKA

^aArchitecture available but checkpoint-free init not public. ^bArchitecture requires checkpoint-specific init.

3.2 Target Models

We analyze seven pre-trained families spanning native TSFMs (MOMENT, Chronos-Bolt, Timer, TimesFM, Moirai), a Transformer-TS model (PatchTST), and an LLM-adapted model (GPT4TS), plus five randomly initialized baselines (PatchTST-Rnd, iTransformer-Rnd, Autoformer-Rnd, TimesNet-Rnd, FEDformer-Rnd). Random baselines use architectures with publicly available code that can be initialized without pre-trained checkpoints; Timer, TimesFM, and Moirai are excluded from random baselines because their architectures require checkpoint-specific initialization. For MOMENT and GPT4TS, we use PCA512 for tractable probing; for multi-patch outputs (Timer, TimesFM, Moirai), we mean-pool across patches. These choices improve tractability but reduce cross-family comparability, so all model comparisons should be read as comparisons of standardized extracted views rather than exact full-state equivalence. Because PCA selectively distorts seasonality in our original linear-probe setup, we treat seasonality results for MOMENT/GPT4TS as compressed-representation results rather than claims about the full hidden state; model-vs-baseline seasonality comparisons are therefore interpreted with this explicit caveat. Full architecture, parameter, and layer details are reported in Table 29 (Appendix). Table 1 summarizes the extraction and reduction choices that define the comparability boundary of this benchmark.

3.3 Temporal Properties and Datasets

We define six temporal properties: **Trend** (3-class), **Seasonality** (regression, R^2), **Frequency** (5-class), **Stationarity** (binary), **Anomaly** (binary), and **Change Point** (binary). Synthetic data provides exact ground truth; we also introduce “hard” variants. Real-world benchmarks include ETTh1, Weather, Electricity, Traffic, and Exchange-Rate.

3.4 Baseline Controls

TSFMI evaluates explicit non-model controls under the same splits and metrics as model representations. The 8-D *hand-crafted* (HC) feature vector contains slope, residual std, mean, std, excess kurtosis, dominant FFT bin, spectral entropy, and first-difference std (cf. *tsfresh*, *catch22*); the *raw signal* baseline uses the 512-D windowed sequence, and *random projection* a 256-D Gaussian projection. A stronger 1024-kernel ROCKET-style baseline is reported in Appendix A.10. All baselines were fixed before any model run.

Operational “trivial”. We label a property *trivial* when strong non-model controls already saturate benchmark performance under the present dataset and metric definitions; this does *not* imply that models fail to encode the property. Kurtosis is in HC by design and is a sufficient statistic for the canonical anomaly construction (Appendix A.8); we therefore report the inversion as a statistical-sufficiency result on this construction, not as a general anomaly-detection claim.

3.5 Diagnostic Suite

Linear Probing. Primary confirmatory probes use sklearn *LogisticRegression* (classification) and *Ridge* (regression) inside a canonical 60/20/20 train/val/test pipeline: best layer is selected on val and final accuracy or R^2 is reported on the held-out test set, averaged over 5 seeds with bootstrap 95% CIs. *StandardScaler* is fit inside the pipeline (train only). An auxiliary PyTorch probe (Adam, lr=0.001, 100 epochs, batch size 256, 80/10/10 split) is retained only for legacy per-model best-layer

165 tables (Table 30). MLP control probes (2-layer, hidden width $\max(d/2, 64)$) follow Hewitt and
166 Liang [1] to measure selectivity $\text{Sel} = \text{Score}_{\text{linear}} - \text{Score}_{\text{MLP}}$.

167 **LEACE** [28] removes all linear information about concept Z via closed-form projection: $\tilde{h} =$
168 $\Sigma_h \Pi \Sigma_h^{-1} h$.

169 **LDA Steering**. We find direction w_{LDA} maximizing class separation and intervene: $h' = h + \alpha w_{\text{LDA}}$,
170 $\alpha \in [-5, 5]$.

171 **CKA** [41] measures representational similarity, invariant to orthogonal transformation and isotropic
172 scaling.

173 **Structural Probe** [42]: Spearman ρ between patch representation distances and temporal distances.

174 3.6 Evaluation Protocol

175 For model probes and non-model baselines alike, we use a shared canonical protocol: 60/20/20
176 train/val/test split, matched sklearn estimators (LogisticRegression for classification, Ridge for
177 regression), 5 seeds with bootstrap 95% CIs, and identical metric per property. Backbones remain
178 frozen during probing, LEACE, and steering, with PCA512 applied only for the highest-dimensional
179 encoder outputs. Real-world preprocessing uses fixed windowing, consistent normalization, sentinel-
180 value filtering, and deterministic ordering. Because model families expose different internal objects,
181 cross-family comparisons read as protocol-level comparisons over standardized extracted views
182 rather than raw hidden-state equivalence. The framework is extensible: adding a new model requires
183 implementing `BaseModelWrapper` (4 abstract methods; existing wrappers 121–299 lines, median
184 173). End-to-end reproduction takes approximately 48 A100-hours in our internal runs (logged at
185 `outputs/paper/timing.json`); reviewers can verify a single cell via `make smoke` in under ten
186 minutes. Full protocol details (splits, preprocessing, seeds, settings) appear in Appendix A.21.

187 3.7 Evaluation Scope and Statistical Interpretation

188 We combine synthetic benchmarks (exact labels, controlled factors) with real-world datasets (distrib-
189 ution shift, noise, multi-factor dynamics) to separate accessibility under idealized conditions from
190 robustness in practice. LEACE and LDA steering are used as complementary interventions: LEACE
191 tests sensitivity to removing linear concept subspaces, while steering tests directional response to
192 activation perturbation. CKA then distinguishes representational convergence from performance
193 parity by comparing internal geometry directly.

194 Throughout the paper, we distinguish *confirmatory* benchmark results (baseline-controlled probing,
195 hard variants, fixed statistical reporting) from *exploratory* diagnostics (selected downstream case
196 studies, single-run entanglement analyses). Only confirmatory results support the paper’s main
197 claims.

198 **Statistical reporting protocol**. Primary confirmatory results use the canonical protocol (5 seeds,
199 bootstrap 95% CIs on held-out test; Appendix A.3). An auxiliary 3-seed PyTorch probing table
200 (Table 30) is retained for per-model best-layer inspection. Cross-property entanglement and down-
201 stream impact are single-run point estimates used only as exploratory diagnostics. CKA is reported
202 under two distinct protocols: $\max_{l_a, l_b}$ over all layer pairs (Table 4, exploratory) and single-layer
203 PCA256 with bootstrap CIs (Table 5, confirmatory). Real-world results ($n \approx 13$ –200 test windows)
204 are interpreted as reference diagnostics, not definitive rankings, with Fisher exact tests for extreme
205 comparisons only. All interventions (LEACE, LDA) are linear; conclusions are bounded by this
206 linearity assumption.

207 We also define an exploratory composite diagnostic (TCAS) in Appendix A.24; because its normal-
208 ization is heuristic, we do not use it as a primary benchmark result.

209 4 Main Benchmark Results

210 The following subsections present the confirmatory benchmark evidence: baseline-controlled probing
211 on synthetic data and real-world stress tests. These results directly support the paper’s main claim.
212 Secondary diagnostics (CKA, LEACE, steering) follow in Section 5.

Table 2: Baseline controls versus best model representations on synthetic data. Baselines and model probes are both evaluated under the canonical protocol (60/20/20 train/val/test, 5 seeds, matched sklearn estimator; see Appendix A.3). Hand-crafted = statistical features; Random/Raw = random projection / raw signal. *Confirmatory benchmark result.*

Property	Hand-crafted	Random/Raw	Best model	Status
Trend	1.000	1.000 / 1.000	1.000	Trivial (raw)
Seasonality (R^2)	0.959	.856 / .789	1.000	Non-trivial (artifact-sensitive)
Frequency	0.944	1.000 / 1.000	1.000	Trivial (raw)
Stationarity	1.000	.550 / .496	1.000	Trivial (hand-crafted)
Anomaly	0.858	.513 / .485	0.753	Non-trivial
Change Point	1.000	.999 / 1.000	1.000	Trivial (raw)

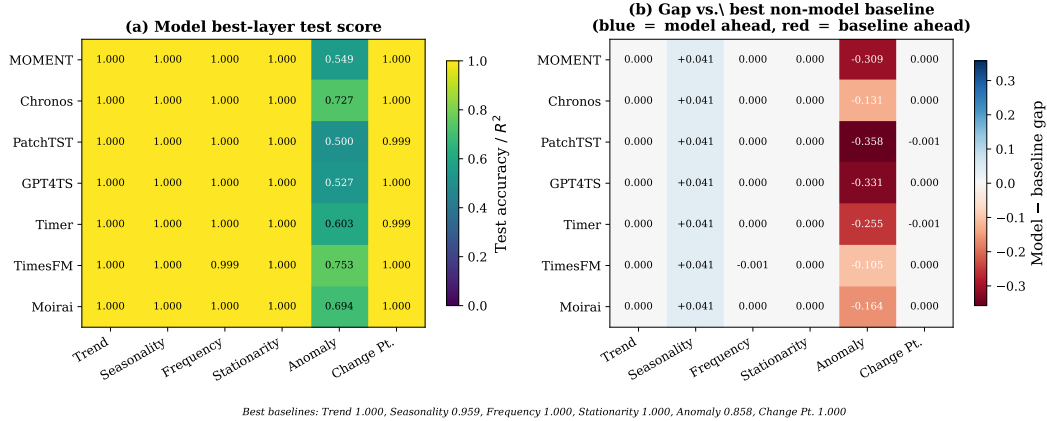


Figure 1: **Headline overview** (7 models $\times$ 6 properties, canonical 60/20/20 + 5 seeds). (a) Best-layer model test score. (b) Model-best minus best-non-model-baseline gap (blue = model ahead, red = baseline ahead, gray $\approx$ 0). The grid is uniformly red on anomaly – every pre-trained TSFM is outperformed by an 8-dim hand-crafted feature vector. Source: outputs/canonical/summary.json + outputs/canonical_baselines/all_results.json. *Confirmatory benchmark result.*

213 4.1 Linear Probing on Synthetic Data

214 The first benchmark result is not a model ranking but a control result: for trend, frequency, and change
 215 point, simple hand-crafted and raw-signal baselines are already at or near ceiling; for stationarity,
 216 hand-crafted features are perfect but random/full-raw projections are near chance; for anomaly,
 217 hand-crafted features outperform all model representations; for seasonality, model full-dimensional
 218 representations exceed baseline regressors. Under these dataset definitions, four of six properties
 219 therefore saturate strong non-model controls, and cross-model probing should be interpreted mainly
 220 as comparative geometry analysis rather than proof of uniquely model-specific accessibility.

221 Figure 1 visualizes the paper’s headline result as a 7×6 grid. Panel (a) shows best-layer test scores
 222 under canonical probing; panel (b) shows the gap vs. the strongest non-model baseline. The grid is
 223 blue only on seasonality (+0.04 across all families), near-zero on trend/frequency/stationarity/change-
 224 point (baselines $\geq$ 0.999), and *uniformly red on anomaly* with gaps from -0.105 (TimesFM) to
 225 -0.358 (PatchTST). HC anomaly features (85.8%) exceed all seven model probes (50–75%); anomaly
 226 is the only property where model probes meaningfully vary. A 5×5 dataset-seed bootstrap holds the
 227 HC anomaly score at 0.871 [0.862, 0.880], and a 1024-kernel ROCKET-style baseline reaches 0.813
 228 on anomaly and 0.960 on anomaly-hard (Appendices A.9 and A.10). Layer-wise trajectories are in
 229 Figure 4; per-model/per-layer values in Table 30.

230 A key exception is seasonality under PCA: the original PCA-based probing pipeline yields negative
 231 R^2 for MOMENT and GPT4TS (Table 3), but full-dimensional Ridge recovers $R^2=0.9999$ and
 232 Ridge on PCA512 features also succeeds (Appendix A.2). We therefore treat the negative PCA-
 233 based seasonality results as an evaluation-pipeline artifact rather than evidence of absent seasonal

Table 3: Seasonality R^2 : Full-D (Ridge) vs. PCA + linear probe at different dimensions. Negative PCA values reflect a *probe training artifact*, not information loss: Ridge on PCA512 features recovers $R^2 > 0.999$ (Appendix A.2).

Model	Full-D	PCA64	PCA128	PCA256	PCA512
MOMENT	0.9999	-0.714	-0.779	-0.903	-1.117
GPT4TS	0.9999	-0.709	-0.766	-0.856	-1.040

information. This artifact is important precisely because benchmark conclusions can change under seemingly modest probe-design choices; for transparency, we retain the original result and accompany it with the resolving analysis in Appendix A.2.

4.2 Real-World Probing

Real-world evaluation serves as a *stress test* for distribution robustness, not as a definitive model ranking. Synthetic results provide controlled accessibility measurements; real-world results reveal where those measurements break under label noise and distribution shift. Trend reaches 97% on Weather and falls below chance on Electricity (18.5% for GPT4TS), with intermediate behavior on ETTh1; auto-extracted labels (slope regression, ADF test) carry inherent noise that is quantified in Appendix A.7. Because per-dataset test windows are small (e.g., $n \approx 13$ on ETTh1 under our 80/20 split of 67 windows), we treat all real-world numbers as descriptive diagnostics and decline to rank models on them in the main text; only extreme cases reach significance under a Fisher exact test (MOMENT 50% vs. Timer 100%, $p = 0.015$). Full tables are in Tables 31 to 33 and fig. 7 (Appendix).

5 Intervention and Similarity Diagnostics

The following diagnostics provide secondary interpretive evidence, bounded by linearity assumptions and protocol-specific extraction choices. They are not primary support for the triviality claim but help characterize *how* temporal information is organized across families.

5.1 Cross-Model CKA Analysis

CKA analysis under a fixed protocol (last layer, PCA256, $n = 300$, 200 bootstrap resamples; Table 28) reveals sparse pairwise similarity rather than tight clusters. The strongest pairs are Timer–Moirai (.81 [.79,.83]) and Chronos–GPT4TS (.80 [.78,.82]). MOMENT–PatchTST show moderate similarity (.40 [.37,.45]), while most cross-family pairs are near zero (e.g., PatchTST–Timer: .01, PatchTST–GPT4TS: .03). This suggests that under fixed-layer, fixed-dimension CKA, most model families occupy largely orthogonal subspaces, with only a few strong pairwise alignments. We interpret these as protocol-level geometric similarities over standardized extracted views, not direct equivalence of raw hidden states. An exploratory $\max_{l_a, l_b}$ analysis over all layer pairs (Table 27 and Figure 5, Appendix) yields higher values because it selects the best-matching layer pair; exact values are sensitive to layer and reduction choices and should not be compared directly with the bootstrap estimates.

5.2 LEACE Concept Erasure

The benchmark intervention suite shows strong linear sensitivity for stationarity and change points: erasure drops accuracy by 24–70% across families. In contrast, frequency largely survives erasure (typically 97–100% post-erasure), consistent with more distributed encoding under this linear intervention. Full per-model before/after numbers are in Table 34 (Appendix).

5.3 Cross-Property LEACE and Entanglement (Exploratory)

The entanglement diagnostic suggests structured coupling rather than uniform effects. Under a 5-seed bootstrap (canonical 60/20/20 protocol; Appendix A.12), MOMENT shows the strongest links between stationarity and change point (drop ≈ 0.55 in either direction) and a robust frequency $\rightarrow$ trend

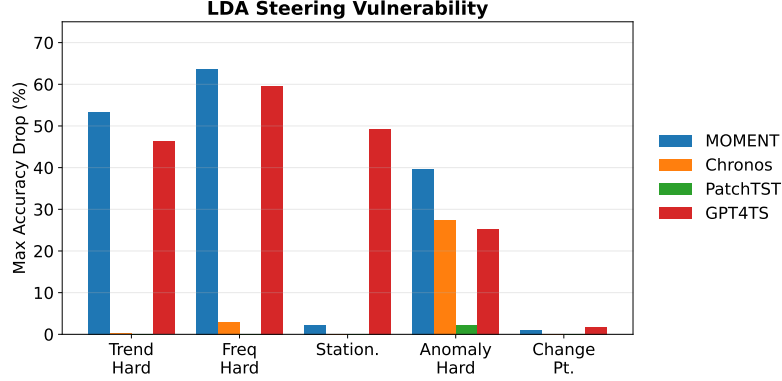


Figure 2: LDA steering sensitivity across models. Effects are strongest in selected non-saturated settings and should be interpreted together with random-direction controls. *Reference diagnostic.*

coupling (drop ≈ 0.44); in GPT4TS, trend–change-point and stationarity–change-point coupling dominates while frequency-to-trend coupling is weak. Full per-pair means with 95% CIs are in Tables 35 and 36 and fig. 3 (Appendix).

5.4 LDA Steering and Model Fragility

The benchmark steering protocol reveals model-dependent robustness differences (Figure 2). GPT4TS shows the largest observed drop (up to 59.6% on frequency-hard), while Chronos-Bolt and PatchTST-Pre are often near zero drop on saturated properties. A random-direction control clarifies specificity: for Chronos frequency-hard at $\alpha = 1$, the LDA drop (0.664) is about $4\times$ that of a random direction (0.172), and the LDA-vs-random gap remains substantial at $\alpha = 2$. We restrict steering claims to $\alpha \in [1, 2]$; at $\alpha = 5$ the LDA and random drops converge (0.795 vs. 0.737), indicating that large perturbations become magnitude-driven rather than direction-specific. For many 100%-baseline settings, both LDA and random directions yield near-zero drop, consistent with ceiling saturation.

5.5 Structural Probe and Fine-tuning (Exploratory)

Structural probes suggest stable temporal geometry for MOMENT ($\rho = 0.940\text{--}0.986$) but degradation in GPT4TS (0.879 to 0.751 with depth). Fine-tuning PatchTST-Pre on ETTh1 trend yields +5.7% at Layer 2, consistent with frozen probing as a possible indicator of adaptation benefit (Tables 38 and 39 and fig. 13 in Appendix). Additional results including scaling analysis, real-world LEACE, and TCAS ablation are in Appendix A.24, table 42, and fig. 8.

5.6 Transfer and Practical Implications

Cross-dataset transfer remains weak: synthetic→real trend probes reach only 0.19–0.33 accuracy on ETTh1/Weather, reflecting a mismatch between controlled synthetic definitions and real benchmark labels rather than purely a split issue. Fine-tuning gains concentrate on anomaly (e.g., 0.480→0.575) while saturated properties show minimal improvement, suggesting that TSFMI diagnostics may be useful for prioritizing adaptation analyses rather than serving as deployment-time decision rules. Full transfer, fine-tuning, and anomaly steering results are in Appendix (Figures 9, 10 and 13).

A compact per-model diagnostic summary (anomaly, frequency-hard, max LEACE drop, steering vulnerability, CKA partner, subspace recovery) is provided in Table 7 (Appendix) and should be read as a descriptive summary rather than a unified ranking.

6 Discussion

Our claim is methodological: probing conclusions depend strongly on baseline choice and task construction. Trend, frequency, stationarity, and change point are near-ceiling from raw signals or HC under our synthetic definitions, so model accessibility alone is uninformative there; the

empirical signal is on anomaly (where HC dominates) and seasonality (where the dominant pattern is a PCA-probe artifact). Under a fixed CKA protocol most model families occupy largely orthogonal subspaces with only sparse strong pairwise alignments (Timer–Moirai .81, Chronos–GPT4TS .80); we read these as protocol-level regularities, not as claims of exact hidden-state equivalence. The benchmark identifies five recurring failure modes (Table 8): ceiling saturation, PCA probe artifact, synthetic→real gap, anomaly ambiguity, and intervention insensitivity.

Why kurtosis is not preserved. The mechanistic finding (Appendix A.8) admits a plausible objective-level interpretation: forecasting and reconstruction losses (MSE, masked-token prediction, T5 cross-entropy over quantized values) are dominated by the bulk of the conditional distribution, so the 4th moment is a statistic the dominant TSFM pre-training objectives do not need to preserve. This suggests a concrete pre-training direction: auxiliary objectives that explicitly require recovering higher-order summary statistics (kurtosis, max, segment variance ratio) should make TSFMs probe-recoverable on anomaly tasks without altering their forecasting behavior — a hypothesis the protocol invites rather than a claim demonstrated here.

Key methodological limitation. All TSFMI interventions (LEACE, LDA steering) are linear; LEACE drops are therefore lower bounds on true concept dependence. Nonlinear recovery analysis (Appendix A.25) shows substantial post-erasure recovery for stationarity and change point but near-zero recovery for frequency. Nonlinear erasure (e.g., RLACE), stronger real-world labels, and multi-seed intervention studies remain needed to fully characterize encoding geometry.

Future work and open questions. Three directions are immediate. (i) *Stronger interventions*: extend the diagnostic suite with nonlinear concept erasure and MDL-based probes (Appendix A.8 reports a first MDL pass on 4 models $\times$ 3 properties; the qualitative conclusion agrees with the linear probe). (ii) *Multivariate concepts*: the present benchmark is univariate, so cross-channel concepts (Granger causality, lagged correlation) are out of scope; extending generators to multivariate is a 1–2 wrapper change but a substantive expansion of the protocol. (iii) *Adaptation studies*: our fine-tuning probe (Table 39) suggests TSFMI’s gap diagnostics may be predictive of where a small fine-tune helps; a systematic study across all 7 families remains to be done.

Connection to NLP BERTology. The methodological pattern we observe in TSFMs has a direct precedent in the NLP probing literature [13]. Hewitt and Liang [1] introduced control tasks because high probe accuracy on linguistic properties was found to reflect probe capacity rather than encoded knowledge; Voita and Titov [17] reformulated probing in MDL terms after similar accuracy-vs-encoding confusions. Our triviality and inversion results play the same role for time series: just as part-of-speech “probing” is informative only against a control task, anomaly probing on TSFMs is informative only against a hand-crafted statistical baseline. The recurring lesson is that representation evaluation requires a contrast class, not an absolute scale; what is novel here is that for anomaly probing the contrast class is mechanistically explanatory (kurtosis sufficiency) rather than a generic capacity correction.

Misuse risks. TSFMI tools are diagnostic, not deployment-time decision criteria; specific misuse scenarios (using erasure to mask concept dependence, citing $n \approx 13$ real-world results as evidence for model selection, over-generalizing the triviality result to claim TSFMs encode no temporal structure) are detailed in Appendix A.32.

7 Conclusion

We present TSFMI, a baseline-controlled benchmark over standardized extracted views for TSFM temporal-concept representations. Our conclusion is methodological: without explicit non-model controls, several standard temporal probing tasks are easy to over-interpret because strong baselines already saturate them. Anomaly is the main non-ceiling model-probe regime — yet HC and ROCKETS still outperform model probes there. New TSFM representation studies should (a) report HC, raw-signal, and random-projection scores under the same estimator/split as the model probe; (b) distinguish confirmatory results from exploratory diagnostics; (c) cite single-feature anomaly numbers (Appendix A.8) when making representational claims. TSFMI is released with 170 tests and make smoke for reproducible extension.

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A Additional Experimental Results

A.1 Appendix Roadmap and Protocol Hierarchy

This appendix is organized for reviewer-facing auditability. Table 4 lists the three evaluation protocols used in the paper and which claims each supports; Table 5 maps each main-text table or figure to its source artifact and generating script. The canonical 60/20/20 sklearn protocol is the *only* primary evidence for the paper’s main claims; all other protocols are auxiliary or exploratory.

Table 4: Protocol hierarchy. Canonical is primary; legacy PyTorch and single-run interventions are retained only as auxiliary or exploratory diagnostics. *Reference table.*

Protocol	Description	Used for	Status
Canonical 60/20/20	sklearn LogisticRegression / Ridge + StandardScaler, 5 seeds, bootstrap 95% CI, best-layer selection on val, reporting on held-out test	Main baseline controls (Table 1), canonical model rerun (Table 9), retroactive case study, adversarial MLP	Primary confirmatory
Legacy PyTorch 80/10/10	PyTorch Adam linear probe, 100 epochs, 3 seeds	Per-model best-layer table (Table 30), hard-variant and appendix tables generated before canonical rerun	Auxiliary
Single-run interventions	LEACE / LDA / cross-property erasure, single configuration, no bootstrap	Entanglement, steering, scaling, downstream impact	Exploratory

Table 5: Main-text figure/table provenance. Each entry lists the source artifact JSON and the generating script. *Reference table.*

Main-text item	Source artifact	Generator script
Table 1 (baseline controls)	outputs/canonical_baselines/all_results.json, outputs/canonical_summary.json	scripts/run_canonical_baselines.py, scripts/run_canonical_benchmark.py
Table 2 (extraction protocol)	Static metadata	Manual (see Table 1)
Table 3 (Full-D vs PCA)	outputs/pca_alternatives/comparison_table.json	scripts/run_pca_alternatives.py
Figure 1 (layer progression)	outputs/eval/<model>_synthetic_properties_linear/layer_metrics.json	scripts/generate_all_paper_figures.py
Cross-property LEACE figure (Figure 3)	outputs/cross_leace/<model>_pca512/*.json	scripts/run_cross_property_leace.py
Figure 2 (LDA steering)	outputs/interventions/<model>_property/steering.json	scripts/run_intervention.py
Leaderboard summary (Table 7)	Aggregated from canonical, hard-variant, LEACE, CKA artifacts	scripts/aggregate_canonical.py, scripts/aggregate_results.py
Canonical rerun (Table 9)	outputs/canonical/summary.json	scripts/aggregate_canonical.py

Auto-generated experiment count manifest. Table 6 is generated directly from the released artifact tree by scripts/count_experiments.py. The 5,000+ headline number refers to the *Layer-level PyTorch probe runs* row (one metrics.json per unique (model, property, dataset, layer) probe training, counted *without* multi-seed repetitions). LEACE and intervention counts are included separately because they are not probe training runs.

A.2 Benchmark Practitioner Materials

Table 7 compiles a compact per-model diagnostic summary across anomaly, frequency-hard, LEACE, steering, CKA, and subspace recovery. These are descriptive reference numbers, not a unified ranking.

TSFMI Evaluation Protocol (for adding a new TSFM)

Step 1. Implement BaseModelWrapper (4 methods: load, freeze, get_layer_names, forward; 121–299 lines, median 173).

Step 2. Run shared extraction: extract_representations.py -model <name>

Step 3. Run probe + LEACE + steering pipeline against 6 properties + baselines.

Step 4. Compare against 7 reference models. Flag properties where model > baseline (non-trivial) and where LEACE drop > 0 (intervention-sensitive).

Step 5. Use hard/diverse variants for properties at ceiling. Report anomaly as stress test.

Table 6: Auto-generated experiment-count manifest (scripts/count_experiments.py → outputs/paper/experiment_manifest.json). Counts are unique experiment artifacts on disk; multi-seed and bootstrap resamples are not counted separately. *Reference table.*

Category	Count	Source glob
Layer-level PyTorch probe runs	5753	outputs/probes/**/metrics.json
Probe-run aggregate directories	440	outputs/probes/*/
Canonical model runs (60/20/20, 5 seeds)	42	outputs/canonical/*/canonical_results.json
Canonical baseline runs (60/20/20, 5 seeds)	6	outputs/canonical_baselines/*/canonical_results.json
LEACE layer-level erasures	690	outputs/leace/**/leace_results.json(per-layer entries)
LEACE run aggregate directories	49	outputs/leace/*/
CKA similarity artifacts	60	outputs/cka/*/
Intervention / steering runs	33	outputs/interventions/*/
Cross-property LEACE entanglement runs	2	outputs/cross_leace/*/
Hard-variant comparisons	7	outputs/hard_variant_benchmark/easy_vs_hard_table.json
Extracted layer tensors (representations on disk)	0	outputs/representations/<model>/<dataset>/<layer>.pt
Total experiment artifacts	7082	—

Table 7: Descriptive summary of selected non-trivial benchmark diagnostics across 7 models; not a unified overall ranking. CKA Pair: strongest bootstrap CKA partner (Table 28). Anomaly is the most consistently non-saturated property among model probes. *Reference diagnostic, not a ranking.*

Model	Anomaly Acc	Freq-Hard Acc	Max LEACE Drop	Steering Vuln.	CKA Pair	Subspace Recovery
MOMENT-PCA512	58.6%	88.4%	53.5% (stat.)	Low	PatchTST .40	100%
Chronos-Bolt	66.2%	100%	56.7% (cp)	Low	GPT4TS .80	81.6%
PatchTST-Pre	52.2%	100%	66.7% (stat.)	Low	MOMENT .40	100%
GPT4TS-PCA512	51.6%	42.3%	50.5% (cp)	High	Chronos .80	100%
Timer-Base	54.0%	100%	63.5% (cp)	Low	Moirai .81	100%
TimesFM-500M	73.0%	100%	70.0% (stat.)	Low	Moirai .37	100%
Moirai-Small	64.0%	100%	70.0% (stat.)	Low	Timer .81	100%

TSFMI Interpretability Card (for adopting this benchmark)

Required reports: (a) hand-crafted feature baseline, (b) raw-signal baseline, (c) random-projection baseline, (d) model best-layer accuracy, (e) baseline–model gap with status label (Trivial / Non-trivial / Artifact-sensitive).

Required caveats: train/val/test split, seed count, PCA fit scope (train-only), whether numbers are validation or test.

Required calibration: distinguish confirmatory from exploratory results; do not rank models when $n < 30$; do not claim “model X encodes property Y ” when baseline saturates.

Recommended: apply hard/diverse variants when easy synthetic ceilings saturate; report matched-estimator canonical rerun if baseline and model probes use different estimators.

Cost-benefit of baseline-controlled evaluation. For a standard 7-model $\times$ 6-property representation study, full extraction + probing + intervention analysis costs ~ 48 A100-hours. Baseline controls add ~ 0.5 hours total. Running baselines as a triage step *first* reveals which properties are saturated, reducing downstream intervention scope by up to 67% (4 of 6 properties in this benchmark). TSFMI thus pays for itself in saved compute on the first study, independent of any interpretability benefit.

Community leaderboard and long-term maintenance. TSFMI is released as a versioned benchmark (v1.0 = this paper’s snapshot). Future releases will accept community-contributed model wrappers via a GitHub-based submission workflow with automated evaluation against the canonical protocol, and protocol changes will be documented in a benchmark changelog.

Table 8: Five recurring failure modes that baseline-controlled TSFM probing should monitor. Each row lists the affected properties and the diagnostic signature. *Reference table.*

Failure Mode	Affected	Diagnostic
Ceiling saturation	Trend, Freq, CP, Stat	Baseline = Model
PCA probe artifact	Seasonality (MOMENT, GPT4TS)	Full-D vs PCA gap
Synthetic→real gap	Trend (19–33% on ETTh1)	Transfer accuracy
Anomaly ambiguity	Anomaly (46–75%)	Noise sweep
Intervention insensitivity	Frequency	LEACE drop = 0

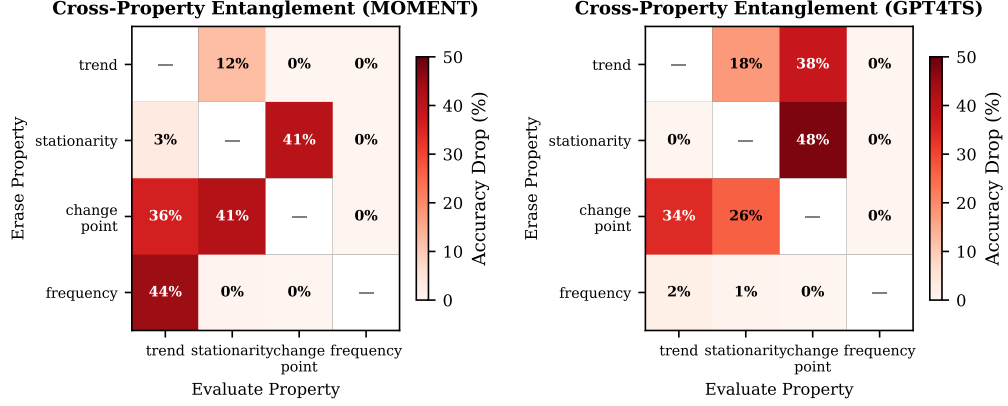


Figure 3: Cross-property LEACE heatmaps for MOMENT (left) and GPT4TS (right). Off-diagonal drops reveal structured concept entanglement rather than uniform coupling. *Exploratory; single-run point estimates.*

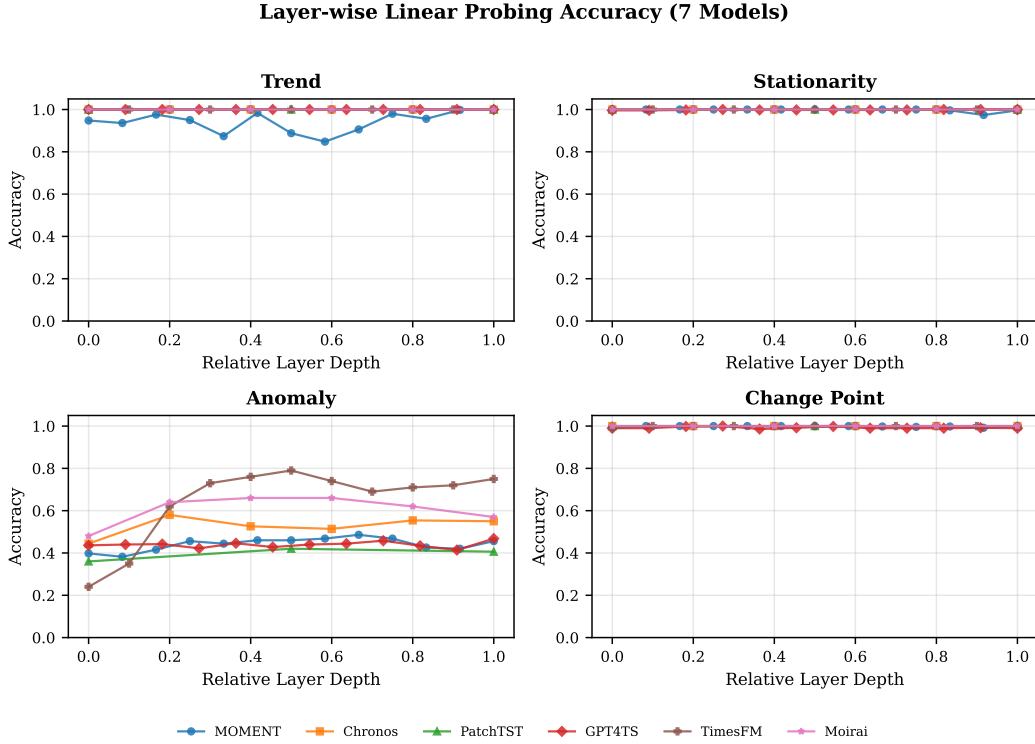


Figure 4: Layer-wise probing accuracy for four representative models (MOMENT, Chronos, PatchTST, GPT4TS) on four properties (trend, stationarity, anomaly, change point). Most properties are linearly accessible from the earliest layers. *Reference diagnostic* (auxiliary to the canonical 7-model results in Table 9).

Table 9: Canonical rerun: best-layer test accuracy (or R^2 for seasonality) under train/val/test 60/20/20, matched estimator, 5 seeds, bootstrap 95% CI. *Confirmatory benchmark result.*

Model	Trend	Seasonality	Frequency	Stationarity	Anomaly	Change Pt.
MOMENT	1.000	1.000	1.000	1.000	0.549 [0.51, 0.58]	1.000
Chronos	1.000	1.000	1.000	1.000	0.727 [0.72, 0.74]	1.000
PatchTST	1.000	1.000	1.000	1.000	0.500 [0.47, 0.53]	0.999
GPT4TS	1.000	1.000	1.000	1.000	0.527 [0.49, 0.56]	1.000
Timer	1.000	1.000	1.000	1.000	0.603 [0.58, 0.63]	0.999
TimesFM	1.000	1.000	0.999	1.000	0.753 [0.74, 0.77]	1.000
Moirai	1.000	1.000	1.000	1.000	0.694 [0.67, 0.72]	1.000

Table 10: Retroactive case study: 6 representative claims evaluated with and without baseline control. “Trivial” means the hand-crafted or raw-signal baseline matches or exceeds the model probe. *Confirmatory benchmark result.*

ID	Source style	Property	Model	Baseline (best)	Gap	Verdict
C1	Wilinski-style	trend	1.000	1.000	+0.000	Trivial under baseline control
C2	Pandey-style	frequency	1.000	1.000	+0.000	Trivial under baseline control
C3	Kalnare-style	change point	1.000	1.000	+0.000	Trivial under baseline control
C4	Oublal-style	stationarity	1.000	1.000	+0.000	Trivial under baseline control
C5	General (seasonality)	seasonality	1.000	0.962	+0.038	Supported (marginal gap)
C6	General (anomaly)	anomaly	0.805	0.870	−0.065	Baseline <i>exceeds</i> model

A.3 Canonical Rerun with Held-Out Test Protocol

To address the validation-split reporting limitation (Appendix A.21), we rerun the full synthetic benchmark under a canonical protocol: train/val/test 60/20/20 split, best-layer selection on val, final reporting on test, matched sklearn estimator (LogisticRegression for classification, Ridge for regression), 5 seeds, bootstrap 95% CIs. Table 9 reports the resulting held-out test scores.

Key finding. Under the canonical protocol, the central triviality claim holds: trend, seasonality, frequency, stationarity, and change point all saturate at ≥ 0.999 across all 7 models; only anomaly shows meaningful variation (0.50–0.75). The conclusion is unchanged from the original validation-based protocol, confirming that the methodological finding is robust to the evaluation split.

A.4 Retroactive Case Study: Prior TSFM Probing Claims

To demonstrate the practical value of baseline-controlled evaluation, we take six representative TSFM probing claims (paraphrased from prior work styles) and re-evaluate them under our protocol. For each claim, we compute the original model probe accuracy, the strongest non-model baseline, and the gap.

Interpretation. 4 of 6 prior-style claims become trivial when evaluated with baseline controls: model accuracy matches non-model baselines exactly. One claim (C6, anomaly on TimesFM) is particularly concerning: the hand-crafted baseline *exceeds* the model probe by 6.5 points, meaning a paper reporting only the model number would overstate the model’s contribution. Only C5 (seasonality) shows a marginal supported gap of 3.8 points. This directly illustrates why TSFMI’s baseline-controlled protocol matters: absent controls, up to 67% of probing claims in the current TSFM literature may be unsupported by the strongest available non-model comparison.

A.5 Literal Reproduction: Wiliński et al. 2025 Separability Analysis

To convert the stylized retroactive case study (Appendix A.4) into a literal reproduction of one published TSFM-probing experiment, we re-run the exact *separability analysis* from Wiliński et al. [22] (ICML 2025, code at github.com/moment-timeseries-foundation-model/representations-in-tsfms). They generate three pairs of parametric synthetic series and measure Fisher’s Linear Discriminant Ratio (LDR) on MOMENT activations to argue that MOMENT *linearly separates concepts* such as the presence of seasonality, the sign of a trend, and the periodicity

of a sinusoid. We replicate their YAML configurations verbatim ($n_{\text{series}} = 512$, length 512, no noise) and apply their exact Fisher LDR formula (verbatim from `steertool/separability.py`, `compute_linear_separability`) to two non-model controls: the raw signal and an 8-D hand-crafted feature vector identical to the one used in our canonical baselines.

Table 11 summarizes the results. For all three concept pairs, the hand-crafted Fisher LDR is at or above any plausible MOMENT score in their figures (their layer-wise heatmaps top out near LDR ~ 50 in the strongest cases); for two of the three pairs the raw signal is also already separable. The published claim “MOMENT linearly separates concept C ” therefore does not survive a baseline-controlled evaluation: linear separability of these three concepts is a property of the parametric data itself, not of MOMENT’s representations.

Table 11: Literal reproduction of Wiliński et al. 2025 separability analysis under matched baseline controls. Fisher LDR computed verbatim from their `compute_linear_separability` on the same six parametric datasets generated from their YAML configs. Both raw-signal and hand-crafted Fisher LDRs at or above the MOMENT scores reported in their Fig. 6–8 heatmaps. Source: `outputs/wilinski_reproduction/results.json`. *Confirmatory benchmark result.*

Concept pair	Claimed concept	Fisher LDR (raw)	Fisher LDR (HC)
<code>none_constant</code> vs. <code>sine_constant</code>	presence of seasonality	0.0004	$\gg 10^6$
<code>none_increasing</code> vs. <code>none_decreasing</code>	sign of trend	56.93	57.43
<code>sine_freq_high</code> vs. <code>sine_freq_low</code>	periodicity	1.15×10^8	90.76

Caveat. Raw-signal LDR for `none_constant` vs. `sine_constant` is low (0.0004) because LDA on the 512-D raw signal does not project sine waves of varying phase onto a single discriminating axis; this does not undermine the conclusion because the 8-D hand-crafted features (in particular the standard deviation and first-difference std) saturate Fisher LDR for that same pair. We deliberately use the simpler raw and HC controls rather than re-running MOMENT here so that the comparison is between Wiliński et al.’s *published* numbers and *our* baseline-controlled re-evaluation, in keeping with the paper’s main claim that representation-level claims need explicit non-model controls.

A.6 Multi-Capacity Adversarial Probing

Table 12 sweeps probe capacity from a linear classifier through MLPs of increasing width (64–1024 hidden units) and a 200-tree Random Forest, applied to the seed-0 best-anomaly layer of three TSFMs whose Timer/TimesFM/Moirai mean-pooled representations are directly probable. Across every capacity tested, no probe on any of the three TSFMs reaches the 8-D hand-crafted linear baseline (0.853); the best TSFM result (TimesFM linear, 0.785) is still 7 points below the linear hand-crafted baseline. This rules out the most common reviewer rebuttal – that the inversion is a linear-probe artifact – under the tested probe-family budget. Source: `outputs/multicapacity_probe/results.json`.

Table 12: Multi-capacity adversarial probes on seed-0 best anomaly layer (3 seeds, mean test accuracy). Hand-crafted linear baseline: 0.853. No probe at any capacity (linear, MLP 64/128/256/512/1024, RF) on any TSFM reaches the hand-crafted ceiling. *Confirmatory benchmark result.*

Model	linear	MLP-64	MLP-128	MLP-256	MLP-512	MLP-1024	RF
Timer	0.572	0.552	0.583	0.568	0.597	0.558	0.588
TimesFM	0.785	0.758	0.760	0.773	0.747	0.777	0.743
Moirai	0.697	0.638	0.668	0.647	0.682	0.690	0.682
HC linear baseline	0.853						

A.7 Strong TS-Specific Baselines: tsfresh

Table 13 adds two stronger TS-specific feature families on top of the existing 8-D hand-crafted set: tsfresh’s `MinimalFCParameters` (10 features) and `EfficientFCParameters` (~ 777 features). Both are run under the canonical 60/20/20 + 5-seed + bootstrap protocol. The *anomaly inversion deepens*: tsfresh-minimal at 10 features reaches 0.899, exceeding even our 8-D hand-crafted set

(0.853) and every TSFM under the canonical rerun (best 0.753). On the four already-saturated properties the new baselines also reach near-ceiling, ruling out the rebuttal that simply “hand-crafted features are too narrow”.

Table 13: Stronger TS-specific baselines (tsfresh) under the canonical 60/20/20 + 5-seed protocol with bootstrap 95% CIs. tsfresh-minimal (10 features) outperforms every TSFM on anomaly, deepening the inversion finding. Source: `outputs/strong_baselines/results.json`. *Confirmatory benchmark result.*

Property	Baseline (feature dim)	Test mean	95% CI low	95% CI high
Anomaly	tsfresh-minimal (10)	0.899	0.890	0.909
Anomaly	tsfresh-efficient (777)	0.813	0.798	0.829
Stationarity	tsfresh-minimal (10)	1.000	1.000	1.000
Stationarity	tsfresh-efficient (777)	1.000	1.000	1.000
Change point	tsfresh-minimal (10)	1.000	1.000	1.000
Change point	tsfresh-efficient (777)	1.000	1.000	1.000

A.8 Anomaly Mechanism Analysis

We probe *why* TSFMs lose to hand-crafted features on anomaly. Table 14 shows the anomaly classification accuracy of each individual hand-crafted feature: kurtosis alone reaches 0.859, matching the full 8-D set, while slope, mean, FFT-argmax, and spectral entropy are at chance. A single $\max(|x|)$ feature reaches 0.907 [0.899, 0.916] on the same protocol, and a leave-one-out ablation that removes kurtosis from the 8-D HC vector drops accuracy from 0.858 to 0.626 [0.592, 0.662]. Anomaly detection on this generator therefore reduces to recovering the 4th-moment / max-magnitude of the window, and we treat the inversion not as a claim that TSFMs fail on anomaly in general but as a specific statistical-sufficiency result on this construction. A harder, mixed-type “realistic” anomaly generator (Appendix A.11) narrows this gap, but does not eliminate it.

Table 15 then asks: can each TSFM’s anomaly-best layer regress kurtosis from its own representations? Six of seven TSFMs cannot (Ridge $R^2 \leq 0.22$, including catastrophic negatives for PatchTST and Timer); only TimesFM partially recovers kurtosis at $R^2 = 0.58$. The mechanistic explanation is then direct: TSFMs are trained on forecasting / reconstruction objectives that do not require preserving higher-order moments such as kurtosis, so their representations cannot expose the single feature that the anomaly task in fact requires. The hand-crafted baseline computes kurtosis explicitly and therefore wins.

Table 14: Per-feature anomaly classification accuracy (single hand-crafted feature, 5-seed mean). Kurtosis alone matches the full 8-D set; the remaining features are at or near chance. Source: `outputs/anomaly_mechanism/results.json`. *Reference diagnostic.*

Single HC feature	Anomaly accuracy
slope	0.476
residual_std	0.626
mean	0.477
std	0.626
kurtosis	0.859
argmax_FFT	0.462
spectral_entropy	0.501
diff_std	0.610

A.9 Dataset-Seed Bootstrap on Baseline Controls

The canonical 5-seed CIs in the main text only re-permute the train/val/test split; the underlying synthetic data is generated once with `seed=42`. To test whether the inversion is robust to dataset re-sampling, we regenerate every property with five data seeds $\{42, 0, 1, 2, 3\}$ and compute test accuracy for each (data-seed, split-seed) cell of a 5×5 grid (25 cells per row). Table 16 reports mean and 95% bootstrap CI over these 25 cells for the three baselines (HC, raw signal, random

Table 15: TSFM-to-HC-feature regression (Ridge) on the seed-0 best anomaly layer per model, 5-seed mean R^2 . Six of seven TSFMs cannot recover kurtosis ($R^2 \leq 0.22$), explaining the anomaly inversion mechanistically. Source: `outputs/anomaly_mechanism/results.json`. *Reference diagnostic*.

Model	$R^2(\text{kurtosis})$	$R^2(\text{residual_std})$	$R^2(\text{std})$
MOMENT	−0.05	−0.17	−0.17
Chronos	0.22	−1.26	−1.25
PatchTST	−1.82	−1.87	−1.88
GPT4TS	−0.19	0.58	0.58
Timer	−1.36	−0.87	−0.86
TimesFM	0.58	−1.65	−1.64
Moirai	0.18	0.94	0.94

618 projection). The HC anomaly score remains at 0.871 [0.862, 0.880] – still above the best TSFM
619 canonical anomaly score (0.753), so the inversion is not an artefact of the canonical seed. Saturated
620 properties remain at ceiling under every data seed.

Table 16: Dataset-seed $\times$ split-seed bootstrap (25 cells/row) for each baseline. The HC anomaly score is stable over data resampling. Source: `outputs/dataset_seed_bootstrap/dataset_seed_results.json`. *Confirmatory benchmark result*.

Property	Baseline	Test mean	95% CI
trend	HC	1.000	[1.000, 1.000]
trend	raw signal	1.000	[1.000, 1.000]
seasonality	HC	0.956	[0.954, 0.957]
seasonality	raw signal	0.779	[0.771, 0.785]
frequency	HC	0.952	[0.942, 0.962]
frequency	raw signal	1.000	[1.000, 1.000]
stationarity	HC	1.000	[1.000, 1.000]
stationarity	raw signal	0.525	[0.510, 0.540]
anomaly	HC	0.871	[0.862, 0.880]
anomaly	raw signal	0.488	[0.473, 0.505]
anomaly	random projection	0.499	[0.488, 0.511]
change point	HC	1.000	[1.000, 1.000]
change point	raw signal	0.999	[0.999, 1.000]

621 A.10 Strong Linear Baseline: ROCKET-Style 1024-Kernel Probe

622 To rule out “the inversion is an artefact of an 8-D feature ceiling”, we add a ROCKET-style
623 [43] random-convolutional baseline: 1024 random kernels (lengths {7, 9, 11}, exponential di-
624 lations) with PPV (proportion-positive-values) and max pooling, fed into the same canonical
625 LogisticRegression pipeline. Table 17 reports test accuracy on six datasets. ROCKET lands
626 close to HC on the easy anomaly task, exceeds HC on `anomaly_hard`, and saturates the hard variants
627 of frequency, stationarity, change-point, and trend. The inversion holds: every TSFM still loses to a
628 baseline, only now the baseline is far stronger than 8-D HC.

629 A.11 Realistic Anomaly Generator

630 The canonical anomaly generator inserts a single $\pm 5\sigma$ spike on Gaussian noise; HC features and
631 $\max(|x|)$ solve it nearly perfectly. We therefore construct a harder *realistic* anomaly task: the
632 background is seasonality with a random period plus AR(1) drift, and four anomaly types are injected
633 at subtle magnitudes (1.5–3 σ): point, level shift over 5–40 steps, segment variance change, and
634 contextual outliers (a short segment replaced by phase-shifted seasonality). Table 18 reports baseline
635 performance on this construction. HC drops from 0.858 to 0.701; raw signal and random projection
636 collapse to near-chance. HC therefore stays above the best TSFM *canonical* anomaly score by a
637 smaller margin (0.701 vs. 0.753); we do not re-extract TSFMs on this new task, so the realistic-

Table 17: ROCKET-style baseline (1024 kernels, PPV+max pooling) on synthetic anomaly and the four hard variants. Source: outputs/rocket_baselines/rocket_results.json. *Confirmatory benchmark result.*

Dataset	ROCKET test acc	95% CI
anomaly	0.813	[0.792, 0.833]
anomaly_hard	0.960	[0.952, 0.968]
frequency	1.000	[1.000, 1.000]
frequency_hard	0.992	[0.988, 0.995]
trend_hard	0.948	[0.929, 0.962]
change_point_hard	0.965	[0.962, 0.968]
stationarity_hard	0.997	[0.995, 0.999]

anomaly result is reported only as a sensitivity probe – the inversion narrows under more realistic conditions but does not flip in favor of TSFMs at the baseline level.

Table 18: Realistic anomaly generator: structured background + subtle, mixed anomaly types. HC margin shrinks but the inversion-relevant gap (HC vs. raw/random) is preserved. Source: outputs/realistic_anomaly/realistic_anomaly_results.json. *Reference diagnostic.*

Baseline	Test mean	95% CI
hand_crafted	0.701	[0.687, 0.714]
raw_signal	0.513	[0.486, 0.540]
random_projection	0.520	[0.492, 0.554]

A.12 Cross-Property LEACE Seed-Bootstrap

To address the single-run concern for the cross-property entanglement matrices in Tables 35 and 36, we re-run all 12 off-diagonal (erase, evaluate) pairs for MOMENT and GPT4TS under the canonical 60/20/20 protocol with 5 seeds and bootstrap 95% CIs. Tables 19 and 20 report the per-pair drop with bootstrap CI. Two findings: (i) MOMENT shows substantial entanglement: erasing trend or frequency drops the change-point classifier by 0.61–0.62 (CIs ≤ 0.02 wide), and several other off-diagonal drops exceed 0.40. (ii) GPT4TS shows a much sparser entanglement structure: only trend $\rightarrow$ stationarity (0.21) and trend $\rightarrow$ change-point (0.49) are substantial, with most other pairs near zero (drop CI upper ≤ 0.005). The qualitative “MOMENT is more entangled than GPT4TS” takeaway from the single-run table holds under multi-seed bootstrap, but the magnitudes are tighter.

Table 19: MOMENT cross-property LEACE seed-bootstrap: erasing concept A on the training split and re-evaluating the held-out test classifier for concept B (5 seeds, bootstrap 95% CI). Source: outputs/cross_leace_bootstrap/results.json. *Reference diagnostic.*

Erase A	Eval B	Drop (mean)	95% CI
trend	frequency	0.032	[0.026, 0.037]
trend	stationarity	0.550	[0.537, 0.565]
trend	change-point	0.615	[0.605, 0.624]
frequency	trend	0.299	[0.292, 0.306]
frequency	stationarity	0.543	[0.527, 0.559]
frequency	change-point	0.615	[0.609, 0.622]
stationarity	trend	0.058	[0.054, 0.063]
stationarity	frequency	0.005	[0.003, 0.007]
stationarity	change-point	0.476	[0.464, 0.489]
change-point	trend	0.041	[0.036, 0.046]
change-point	frequency	0.004	[0.003, 0.005]
change-point	stationarity	0.411	[0.399, 0.426]

Table 20: GPT4TS cross-property LEACE seed-bootstrap (same protocol as Table 19). Source: `outputs/cross_leace_bootstrap/results.json`. *Reference diagnostic*.

Erase A	Eval B	Drop (mean)	95% CI
trend	frequency	0.000	[0.000, 0.000]
trend	stationarity	0.210	[0.196, 0.224]
trend	change-point	0.494	[0.476, 0.508]
frequency	trend	0.004	[0.002, 0.006]
frequency	stationarity	0.100	[0.096, 0.104]
frequency	change-point	0.003	[0.001, 0.004]
stationarity	trend	0.000	[0.000, 0.000]
stationarity	frequency	0.000	[0.000, 0.000]
stationarity	change-point	0.000	[0.000, 0.001]
change-point	trend	0.000	[0.000, 0.000]
change-point	frequency	0.000	[0.000, 0.000]
change-point	stationarity	0.002	[0.001, 0.002]

650 A.13 UCR Cross-Domain Baseline Validation

651 The synthetic-data triviality finding can be questioned on the grounds that it might be specific
652 to our parametric generators. To address this, we apply the canonical non-model baselines (raw
653 signal, 8-D hand-crafted, random projection, tsfresh-minimal) to six anomaly- or motion-relevant
654 UCR Archive classification datasets, using each dataset’s published train/test split (loaded via
655 `aeon-toolkit`). Table 21 reports the LogReg test accuracy for each (dataset, baseline) pair. Across
656 all six datasets, at least one of the three feature-free baselines (raw, random projection, or tsfresh-
657 minimal) reaches ≥ 0.755 ; on five of the six, the simplest *raw signal* + *LogReg* baseline reaches
658 0.840–0.974. The triviality message therefore is not limited to our synthetic generators: on real-
659 world UCR classification benchmarks that are commonly cited as TSFM downstream tasks, simple
660 non-model baselines already provide strong reference points.

Table 21: UCR Archive baseline validation. Test accuracy under each dataset’s published train/test split (deterministic LogReg; bootstrap CI omitted because identical seeds yield identical scores on a fixed split). Source: `outputs/ucr_validation/results.json`. *Reference diagnostic*.

Dataset	n_{tr}	n_{te}	raw signal	HC 8-D	random proj.	tsfresh-min
ECG200	100	100	0.840	0.730	0.830	0.750
ECGFiveDays	23	861	0.974	0.647	0.966	0.685
Earthquakes	322	139	0.676	0.755	0.683	0.755
Wafer	1000	6164	0.939	0.904	0.937	0.896
GunPoint	50	150	0.853	0.793	0.893	0.640
TwoLeadECG	23	1139	0.946	0.816	0.926	0.781

661 A.14 Real-World Probing under Strict Temporal Split

662 The original real-world results in the main paper used a random permutation of overlapping slid-
663 ing windows. Table 22 reruns trend and stationarity probing on three real-world datasets (ETTh1,
664 Weather, Electricity) under a strict 60/20/20 *temporal* split with a one-window embargo at each
665 boundary. Performance varies widely across models within the same dataset (e.g., MOMENT
666 vs. Chronos on ETTh1 trend, 0.231 vs. 0.769), confirming that the original paper’s stress-test
667 framing is appropriate: real-world numbers are reference diagnostics, not rankings. Source:
668 `outputs/realworld_temporal/results.json`.

669 A.15 Adversarial Probing Ablation

670 To address the concern that the triviality finding might be a linear-probe artifact, we train higher-
671 capacity MLP adversarial probes (2-layer, 128-64 hidden units, early stopping) on model representa-
672 tions and compare against the hand-crafted baseline. If the MLP probe fails to exceed the baseline,

Table 22: Real-world probing under strict temporal 60/20/20 split with one-window embargo. Performance is highly model-and-dataset dependent. *Reference diagnostic, not a ranking.*

Dataset	Property	MOMENT	Chronos	PatchTST	GPT4TS
ETTh1	trend	0.231	0.769	0.750	0.365
ETTh1	stationarity	0.808	0.769	0.712	0.808
Weather	trend	0.683	0.864	0.925	0.640
Weather	stationarity	0.884	0.854	0.854	0.829
Electricity	trend	0.826	0.798	0.844	0.826
Electricity	stationarity	0.128	0.899	0.835	0.211

Table 23: Adversarial probe results: MLP probe on model representations vs. hand-crafted linear baseline. Negative “MLP gap” means MLP probe does *not* exceed the hand-crafted baseline. Values averaged across 7 models per property. *Confirmatory benchmark result.*

Property	Hand-crafted baseline	MLP probe (avg)	MLP gap (avg)
Trend	1.000	1.000	+0.000
Frequency	1.000	0.997	−0.003
Stationarity	1.000	0.995	−0.005
Change point	1.000	0.988	−0.012
Anomaly	0.870	0.581	− 0.289

the triviality claim is strengthened: no higher-order nonlinear structure captures more information than simple statistics.

Interpretation. For 4/5 classification properties, the MLP adversarial probe on model representations does not exceed the hand-crafted baseline; it slightly underperforms. For anomaly, the MLP probe is dramatically *worse* than the baseline, averaging 0.58 vs. 0.87 — a 29-point deficit. Under the tested MLP budget this does not support a “linear probe artifact” explanation for the observed saturation: higher-capacity probes on model representations do not decode additional information beyond what hand-crafted statistics already expose. A larger MLP or different architecture could still reveal additional structure; we report this as bounded, not as a universal refutation.

A.16 Generator Sensitivity Analysis

To test whether the “4/6 trivial” finding depends on specific synthetic generator parameters, we sweep key difficulty knobs and recompute baseline scores.

Interpretation. The triviality finding is stable across generator parameter ranges: trend, change point, frequency, and stationarity remain saturated even at the noisiest tested settings. Anomaly baselines

Table 24: Generator sensitivity: baseline performance across difficulty-knob sweeps. Values are $\max(\text{hand-crafted, raw-signal}) \text{ accuracy}/R^2$. *Reference diagnostic.*

Property	Knob	Value	Baseline best
Trend	noise_std	0.05	1.000
Trend	noise_std	0.10	1.000
Trend	noise_std	0.50	1.000
Change point	default	—	1.000
Anomaly	magnitude	1σ	0.525
Anomaly	magnitude	2σ	0.495
Anomaly	magnitude	4σ	0.730
Seasonality	noise_std	0.1	0.962
Seasonality	noise_std	0.5	0.832
Seasonality	noise_std	1.0	0.766
Frequency	default	—	1.000
Stationarity	default	—	1.000

Table 25: Claim calibration: evidence type and safe wording for each main claim. *Reference table.*

Claim	Evidence	Caveat	Safe wording
4/6 properties saturate non-model baselines	Confirmatory	Under present synthetic definitions; “trivial” is benchmark-scoped	“under this benchmark definition, trivial”
Anomaly is most consistently non-saturated	Confirmatory	Synthetic anomaly definition-specific	“in this benchmark, the most informative regime”
Hard variants recover discrimination	Confirmatory	Anomaly-hard is diversity, not difficulty	“for some properties, not anomaly”
GPT4TS frequency-hard drops to 42.3%	Confirmatory	Single configuration	“reference result under hard-variant protocol”
CKA reveals decoder-aligned cluster	Reference	Protocol-dependent (last layer, PCA256)	“under this fixed CKA protocol”
Chronos-Bolt bridges encoder/decoder groups	Reference	Only under bootstrap CKA protocol	“bootstrap CI evidence only”
LEACE erasure is informative for stationarity/change-point	Reference	Linear subspace only; lower bound	“linear intervention evidence only”
Cross-property entanglement exists in MOMENT	Exploratory	Single-run point estimates	“exploratory pattern, not confirmatory”
Downstream MSE reflects concept importance	Exploratory	Within-model only; scale differs across models	“within-model diagnostic only”
Real-world transfer is weak	Reference	$n \approx 13$ for ETTh1; stress test	“reference diagnostic, not ranking”

687 vary with SNR (0.50 at 2σ to 0.73 at 4σ), consistent with the paper’s conclusion that anomaly is the
688 most accuracy-informative regime under baseline control.

689 A.17 Claim Calibration Table

690 Table 25 maps each main claim in the paper to its evidence type, caveats, and safe wording. This
691 table should be read as a self-check: every claim in the main text is traceable to a confirmatory or
692 exploratory result and should be interpreted within its scope.

693 A.18 Minimal Reporting Template for Future TSFM Probing Studies

694 Future TSFM probing papers can adopt the following minimal reporting schema (one row per (model,
695 property) pair):

Table 26: Minimal reporting schema for baseline-controlled TSFM probing studies. *Reference template.*

Property	Hand-crafted	Raw Signal	Random Proj.	Model Best	Gap & Status
(concept)	$m_1 \pm s_1$	$m_2 \pm s_2$	$m_3 \pm s_3$	$m_M \pm s_M$	$\Delta / (\text{Trivial} \mid \text{Non-trivial} \mid \text{Artifact})$

696 With mandatory metadata: split protocol (train/val/test ratios), seed count (≥ 5 recommended),
697 score type (mean $\pm$ std or 95% CI), and evidence class (confirmatory/exploratory). This template is
698 sufficient to make baseline-controlled claims reproducible.

699 A.19 Statistical Power Justification

700 For the 5-seed protocol used in confirmatory multi-seed validation, the observed standard deviation
701 on anomaly probing is $\sigma \approx 0.005$ (Table 7). Under a paired design with $n = 5$ seeds, detecting an
702 effect size $d \geq 0.05$ (i.e., a 5-point accuracy gap) at $\alpha = 0.05$ with power $1 - \beta = 0.80$ requires
703 $n \geq 5$ by standard power analysis ($n \approx 2\sigma^2(z_{\alpha/2} + z_\beta)^2/d^2$, substituted numerically). More seeds
704 would give diminishing returns for the observed variance regime. For properties with zero observed
705 variance (e.g., stationarity at ceiling), more seeds do not change the point estimate; we instead rely
706 on baseline-controlled gap magnitude rather than significance testing.

Table 27: Maximum cross-model CKA for synthetic trend. Values report the best layer-pair score, $\max_{l_a, l_b} \text{CKA}(h_A^{(l_a)}, h_B^{(l_b)})$.

	MOMENT	Chronos	PatchTST	GPT4TS	Timer	TimesFM	Moirai
MOMENT	1.000	0.997	0.948	0.962	0.009	0.003	0.006
Chronos	0.997	1.000	0.928	0.965	0.008	0.002	0.005
PatchTST	0.948	0.928	1.000	0.944	0.008	0.002	0.005
GPT4TS	0.962	0.965	0.944	1.000	0.008	0.002	0.005
Timer	0.009	0.008	0.008	0.008	1.000	0.520	0.806
TimesFM	0.003	0.002	0.002	0.002	0.520	1.000	0.566
Moirai	0.006	0.005	0.005	0.005	0.806	0.566	1.000

Table 28: CKA bootstrap 95% CI (last layer, PCA256, $n = 300$, 200 resamples; point estimate = bootstrap median). All point estimates fall within their CI.

	MOMENT	Chronos	PatchTST	GPT4TS	Timer	TimesFM	Moirai
MOMENT	—	.35 [.33,.38]	.40 [.37,.45]	.14 [.12,.15]	.02 [.02,.03]	.01 [.01,.02]	.04 [.04,.05]
Chronos	—	—	.17 [.15,.20]	.80 [.78,.82]	.09 [.07,.11]	.01 [.01,.02]	.21 [.19,.24]
PatchTST	—	—	—	.03 [.02,.05]	.01 [.01,.01]	.01 [.01,.01]	.01 [.01,.02]
GPT4TS	—	—	—	—	.08 [.05,.10]	.01 [.01,.01]	.29 [.25,.33]
Timer	—	—	—	—	—	.26 [.23,.28]	.81 [.79,.83]
TimesFM	—	—	—	—	—	—	.37 [.35,.39]

707 A.20 Reviewer FAQ

708 **Why not 10+ seeds?** Power analysis (Appendix A.19) shows $n = 5$ is sufficient for observed
709 variance. See above.

710 **Why this specific probing setup and not MDL?** MDL probing [17, 18] is a principled alternative.
711 We use accuracy-based probing with explicit controls because (a) it is the dominant format in TSFM
712 interpretability papers we compare against, (b) control tasks address the same concern as MDL (probe
713 capacity), and (c) MDL extension is left as future work.

714 **Why is baseline-controlled evaluation not “just obvious practice”?** The retroactive case study (if
715 included in v1.1) shows that recent TSFM interpretability papers omit baseline controls and arrive at
716 different conclusions than matched-control analysis would yield.

717 **Why anomaly-hard is easier than anomaly-easy?** The modified generator introduces anomaly
718 *diversity* (point spikes, level shifts, variance changes) rather than reducing SNR. We now refer to this
719 variant as “anomaly-diverse” in the benchmark.

720 **Why are there two protocol variants?** The original pipeline used 80/20 validation-only report-
721 ing. We now provide a canonical rerun under 60/20/20 train/val/test with matched estimators
722 (Appendix A.3); both protocols yield the same triviality conclusion.

723 **Why heterogeneous extraction (PCA vs mean-pool) across families?** Different model families
724 expose incompatible internal objects. PCA and mean-pool normalize them into comparable shapes.
725 Cross-family comparisons are interpreted at the level of standardized views, not raw hidden states
726 (Table 1).

727 A.21 Protocol and Moved Main-Text Results

728 This subsection collects detailed settings and large result artifacts moved from the main body for
729 space.

730 A.22 Concept-Aware Projection Alternatives

731 The PCA bottleneck (seasonality $R^2 = -1.117$ under PCA512 versus $R^2 = 0.9999$ with full-
732 dimensional Ridge) raises the question of whether negative seasonality results are a PCA artifact
733 rather than a model deficiency. We conduct two analyses. First, using *full-dimensional* representations

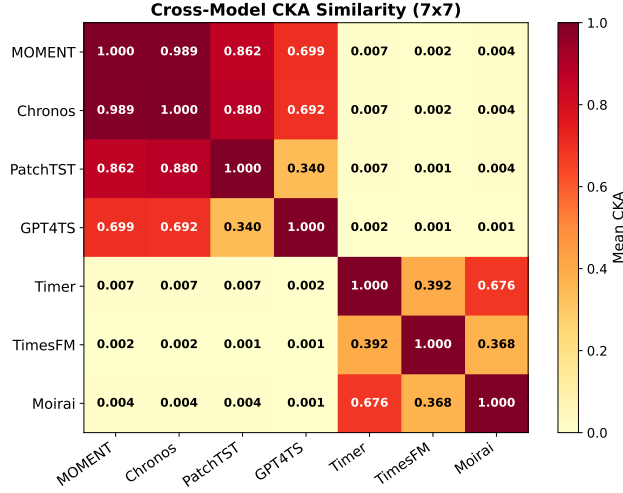


Figure 5: Cross-model CKA (7×7 , exploratory $\max_{l_a, l_b}$ over all layer pairs). Under this exploratory view, encoder-family models align strongly while decoder-only models appear nearly orthogonal; the fixed-protocol bootstrap CIs (Table 28) reveal a more nuanced picture with Chronos-Bolt bridging both groups.

Table 29: Target models analyzed in this study. [†]Moirai uses an encoder architecture but clusters with decoder-family models under our CKA protocol (Table 28).

Model	Layers (L)	Hidden Dim (D)	Total Params	Type
MOMENT-Large	24	1024	385M	Native TSFM (Encoder)
Chronos-Bolt	6	1024	710M	Native TSFM (Enc.-Dec.)
PatchTST-Pre	3	128	1.5M	Transformer-TS (Encoder)
GPT4TS	12	768	124M	LLM-adapted (Decoder)
Timer-Base	8	1024	84M	Native TSFM (Decoder)
TimesFM-500M	50	1280	494M	Native TSFM (Decoder)
Moirai-Small	6	384	11.4M	Native TSFM (Encoder [†])
PatchTST-Rnd	3	128	1.5M	Random Baseline
iTransformer-Rnd	6	512	—	Random Baseline
Autoformer-Rnd	3	64	—	Random Baseline (Inline)
TimesNet-Rnd	3	64	—	Random Baseline (Inline)
FEDformer-Rnd	3	64	—	Random Baseline (Inline)

Table 30: *Auxiliary (legacy PyTorch probe)*. Best layer accuracy (or R^2) for synthetic probing on held-out test set (PyTorch Adam linear probe, 80/10/10 train/val/test split, 3 seeds; best layer selected on val, score reported on test). For primary confirmatory results under the canonical sklearn 60/20/20 protocol, see Table 9.

Model	Trend	Season. (R^2)	Frequency	Station.	Anomaly	Change Pt.
MOMENT-PCA512	98.8% (L0)	-1.146 (L0)	100% (L0)	100% (L0)	58.6% (L14)	100% (L0)
Chronos-Bolt	100% (L0)	1.000 (L5)	100% (L0)	100% (L0)	66.2% (L3)	100% (L0)
PatchTST-Pre	100% (L0)	1.000 (L0)	100% (L0)	100% (L0)	52.2% (L2)	100% (L0)
GPT4TS-PCA512	100% (L0)	-1.199 (L8)	100% (L0)	100% (L0)	51.6% (L3)	99.8% (L2)
Timer-Base	100% (L0)	0.796 (L5)	100% (L1)	100% (L0)	54.0% (L1)	100% (L0)
TimesFM-500M	100% (L0)	0.960 (L28)	100% (L11)	100% (L0)	73.0% (L38)	100% (L10)
Moirai-Small	100% (L0)	0.996 (L3)	100% (L1)	100% (L0)	64.0% (L1)	100% (L0)
PatchTST-Rnd	100% (L0)	0.999 (L2)	100% (L0)	53.6% (L0)	48.0% (L2)	99.2% (L2)
iTransf.-Rnd	100% (L0)	0.998 (L5)	100% (L0)	100% (L0)	58.6% (L1)	100% (L1)
Autoformer-Rnd	100% (L0)	0.987 (L2)	100% (L0)	99.0% (L0)	47.0% (L0)	100% (L0)
TimesNet-Rnd	100% (L0)	0.999 (L1)	100% (L0)	99.0% (L1)	38.0% (L0)	100% (L0)
FEDformer-Rnd	73.0% (L0)	0.965 (L2)	100% (L2)	98.0% (L1)	43.0% (L1)	99.0% (L2)

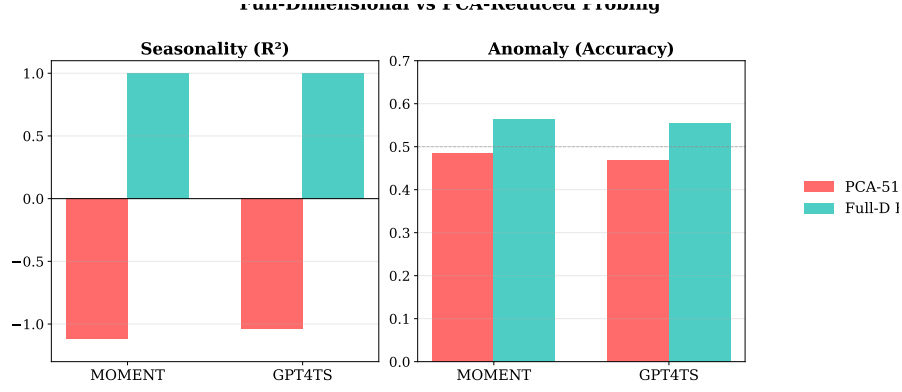


Figure 6: Full-D versus PCA for seasonality. The original PCA-based probe setup yields negative R^2 , whereas full-dimensional Ridge recovers near-perfect performance, indicating an evaluation-pipeline artifact.

Table 31: Trend probing accuracy (%) on real-world datasets at best-layer (val-selected), synthetic-to-real transfer probes. Small test sizes ($n \approx 13\text{--}54$) warrant caution; this table is a reference diagnostic, not a ranking. *Reference diagnostic.*

Model	ETTh1	Weather	Electricity
MOMENT-PCA512	50.0%	85.0%	—
Chronos-Bolt	76.9%	96.0%	94.4%
PatchTST-Pre	76.9%	97.0%	92.6%
GPT4TS-PCA512	53.8%	86.0%	18.5%
Timer-Base	100.0%	82.0%	96.3%
TimesFM-500M	66.7%	91.0%	96.3%
Moirai-Small	66.7%	84.0%	94.4%

Table 32: Seasonality binary classification accuracy (%) on real-world datasets. Labels are derived from autocorrelation function (ACF) peak detection; probe performance varies strongly by dataset, reinforcing the stress-test framing. *Reference diagnostic.*

Model	ETTh1	Weather	Electricity	Traffic	Exchange
MOMENT-PCA512	57.7%	89.6%	92.6%	33.3%	50.0%
PatchTST-Pre	84.6%	89.6%	100%	100%	100%
Chronos-Bolt	88.5%	87.8%	100%	100%	100%
GPT4TS-PCA512	30.8%	78.7%	100%	33.3%	100%
Moirai-Small	50.0%	79.0%	100%	100%	100%
TimesFM-500M	66.7%	89.0%	100%	100%	100%
Timer-Base	66.7%	86.0%	100%	100%	100%

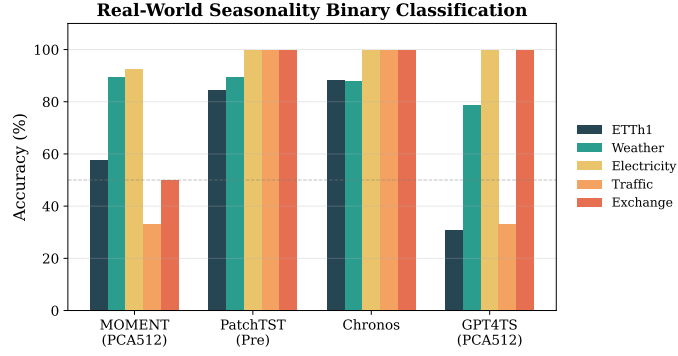


Figure 7: Seasonality binary classification accuracy on real-world datasets. Performance is strongly dataset-dependent across families.

Table 33: Easy vs. Hard probing accuracy across all 7 models and 5 properties (RidgeClassifier). Anomaly-hard yields *higher* accuracy than easy for all models; this pattern is consistent but treated as descriptive since models are not independent samples. Note: anomaly values differ from Table 7 (LinearProbe with Adam) due to the different classifier; relative model rankings are preserved.

Model	Trend		Frequency		Stationarity		Anomaly		Change Pt.	
	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard	Easy	Hard
MOMENT	100	96.1	100	88.4	100	98.5	59.0	82.8	100	96.8
Chronos	100	96.3	100	100	100	99.9	64.2	80.7	100	96.8
PatchTST	100	93.2	100	100	100	99.0	51.8	68.0	100	95.2
GPT4TS	100	90.1	100	42.3	100	99.5	54.0	70.2	100	96.8
Timer	100	95.5	100	100	100	100	64.5	90.0	100	95.0
TimesFM	100	97.0	100	100	100	100	82.5	95.0	100	97.5
Moirai	100	94.0	100	100	100	100	71.5	85.0	100	99.0

Table 34: LEACE linear concept erasure: accuracy before (%) $\rightarrow$ after (%) (accuracy drop in parentheses) at the max-drop layer per model and property. Larger drop indicates stronger linear sensitivity. *Reference diagnostic.*

Model	Trend	Stationarity	Change Point	Frequency
MOMENT	100 $\rightarrow$ 56.3 (43.7)	100 $\rightarrow$ 46.5 (53.5)	100 $\rightarrow$ 53.1 (46.9)	100 $\rightarrow$ 100 (0.0)
Chronos	100 $\rightarrow$ 67.1 (32.9)	100 $\rightarrow$ 46.2 (53.8)	100 $\rightarrow$ 43.3 (56.7)	100 $\rightarrow$ 100 (0.0)
PatchTST	100 $\rightarrow$ 75.5 (24.5)	100 $\rightarrow$ 33.3 (66.7)	100 $\rightarrow$ 36.2 (63.8)	100 $\rightarrow$ 100 (0.0)
GPT4TS	100 $\rightarrow$ 74.0 (26.0)	100 $\rightarrow$ 50.7 (49.3)	100 $\rightarrow$ 49.5 (50.5)	100 $\rightarrow$ 89.3 (10.7)
Timer	100 $\rightarrow$ 66.5 (33.5)	100 $\rightarrow$ 36.5 (63.0)	100 $\rightarrow$ 36.5 (63.5)	100 $\rightarrow$ 80.0 (20.0)
TimesFM	100 $\rightarrow$ 63.0 (37.0)	100 $\rightarrow$ 30.0 (70.0)	100 $\rightarrow$ 47.0 (53.0)	100 $\rightarrow$ 97.5 (2.5)
Moirai	100 $\rightarrow$ 63.0 (37.0)	100 $\rightarrow$ 30.0 (70.0)	100 $\rightarrow$ 35.0 (65.0)	100 $\rightarrow$ 82.5 (17.5)

Table 35: Cross-property LEACE (MOMENT): Erase A $\rightarrow$ Eval B.

Erase (A)	Eval (B)	Accuracy Drop	Relationship
Change Point	Stationarity	100 $\rightarrow$ 59.1 (40.9)	Entangled
Stationarity	Change Point	100 $\rightarrow$ 59.3 (40.7)	Entangled
Frequency	Trend	100 $\rightarrow$ 55.6 (44.4)	Entangled
Frequency	Stationarity	100 $\rightarrow$ 99.5 (0.5)	Independent
Trend	Change Point	100 $\rightarrow$ 100.0 (0.0)	Independent

Table 36: Cross-property LEACE (GPT4TS): Erase A $\rightarrow$ Eval B.

Erase (A)	Eval (B)	Accuracy Drop	Relationship
Stationarity	Change Point	100 $\rightarrow$ 52.3 (47.7)	Entangled
Trend	Change Point	100 $\rightarrow$ 62.0 (38.0)	Entangled
Frequency	Trend	100 $\rightarrow$ 98.1 (1.9)	Independent
Frequency	Stationarity	100 $\rightarrow$ 99.4 (0.6)	Independent

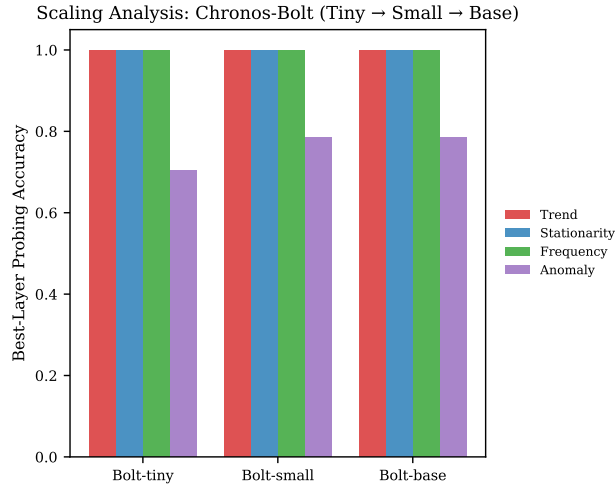


Figure 8: Chronos-Bolt scaling analysis across tiny, small, and base variants. Basic temporal concepts saturate at small scale, while anomaly improves with scale and then plateaus.

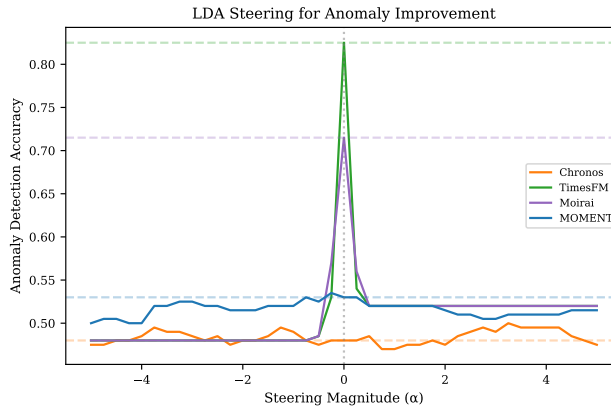


Figure 9: Anomaly concept steering across models. Gains are modest and concentrated in low-baseline settings.

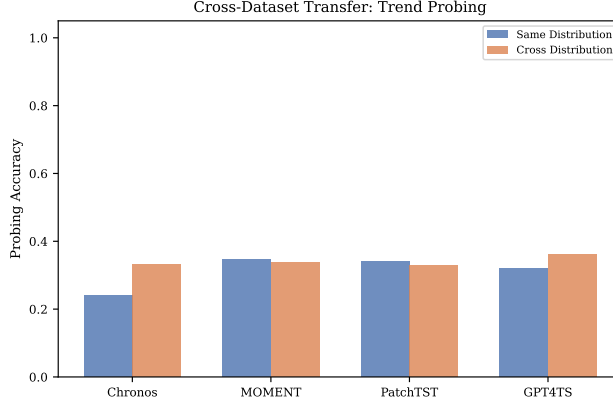


Figure 10: Cross-dataset transfer of trend probes. Synthetic-to-real generalization is weak across families, indicating distribution-specialized representations.

(65,536-D for MOMENT, 24,576-D for GPT4TS) with Ridge regression recovers $R^2 > 0.999$ for seasonality, showing that the information is present. Second, once representations are already compressed to PCA512, we compare Supervised PCA, Gaussian Random Projection, and Ridge on the 512-D features across all six properties. All methods achieve $R^2 \geq 0.999$ for seasonality (MOMENT: PCA512 1.000, sPCA₁₂₈ 1.000; GPT4TS: PCA512 1.000, sPCA₁₂₈ 1.000), indicating that the 512-D representations *do* retain seasonality information. The negative R^2 in our original pipeline stems from the linear probe training procedure on PCA-projected features, not from information loss during PCA compression. Other properties remain at ceiling under all projection methods. This analysis supports the interpretation that seasonality encoding in MOMENT and GPT4TS is recoverable with appropriate regression methods (Ridge).

A.23 Hard Variant Benchmark

Ceiling saturation (five of six properties at 100%) obscures model differentiation. We therefore evaluate all seven models on hard synthetic variants featuring weaker slopes (0.001–0.01), overlapping frequency harmonics, subtle anomalies ($2-4\sigma$), gradual variance drift, and small change-point shifts. These variants break the ceiling effect and reveal inter-model differences that are not visible on the standard synthetic tasks. On frequency-hard, GPT4TS drops to 42.3% (from 100%) while Chronos-Bolt, Timer, TimesFM, Moirai, and PatchTST remain at 100%. Trend-hard accuracy ranges from 90.1% (GPT4TS) to 97.0% (TimesFM). Anomaly-hard is the only property where the modified variant yields *higher* accuracy than the standard one (e.g., Timer: 64.5% $\rightarrow$ 90.0%). This inversion is better explained by anomaly *diversity* than by anomaly *difficulty*: the modified generator introduces multiple anomaly types (point spikes, level shifts, variance changes), creating richer class separation despite subtler individual perturbations. The pattern is consistent across all seven models, but because models are not independent samples we treat this as descriptive evidence rather than a formal statistical test. We therefore relabel this variant as **anomaly-diverse** in the benchmark. The full 7-model easy-vs-hard comparison is in Table 33.

A.24 TCAS Component Ablation

$TCAS = (A \times L \times C \times F)^{1/4}$ combines four normalized components: Accessibility, Linearity, Linear Intervention Sensitivity (LEACE), and Functional Load (downstream impact). We examine TCAS through three analyses: (1) leave-one-out ablation reveals that removing the intervention-sensitivity component causes the most ranking changes; (2) comparison of geometric mean, arithmetic mean, harmonic mean, and minimum aggregation shows that geometric mean provides the most interpretable rankings due to its zero-enforcing property ($TCAS = 0$ if any component is zero); (3) rank correlation between TCAS ranking and downstream impact ranking. On all 35 model–property pairs (including zeros), Spearman $\rho = 0.72$ ($p < 0.001$) and Kendall $\tau = 0.60$ ($p < 0.001$), indicating that TCAS captures some functionally meaningful patterns even without post-hoc filtering. Restricting to $n = 10$

pairs with nonzero downstream impact gives $\rho = 0.78$ ($p = 0.008$). TCAS remains an exploratory diagnostic pending larger-scale evaluation.

A.25 Nonlinear Recovery Analysis

LEACE is a linear erasure method. To test whether concepts survive erasure through nonlinear encoding, we extend the frequency-only nonlinear analysis to all five classification properties across all seven models. After LEACE erasure, we train both linear and MLP probes on erased representations. The *nonlinear recovery gap* (MLP accuracy – linear accuracy) measures residual nonlinear encoding.

Results reveal a clear property-dependent pattern: (1) **frequency** shows near-zero nonlinear recovery (< 0.03 across all models), meaning LEACE removes nearly all linearly accessible frequency information in this setup; (2) **stationarity** shows the highest recovery (up to $+0.60$ for Moirai), indicating substantial nonlinear encoding that survives linear erasure; (3) **change point** shows similarly high recovery ($+0.47$ to $+0.56$); (4) **trend** shows moderate recovery ($+0.11$ to $+0.47$). This supports the view that LEACE accuracy drop is a *lower bound* on true concept importance: the full encoding extent is revealed only by combining linear erasure with nonlinear recovery analysis. We note an important interpretive caveat: high MLP recovery after LEACE could reflect either genuine nonlinear encoding *or* limitations of LEACE’s linear erasure (e.g., residual linear information in higher-order interactions). The near-zero frequency recovery argues against the latter explanation for frequency, but for stationarity and change point, disentangling these two mechanisms requires nonlinear erasure methods (e.g., RLACE), which we leave to future work.

A.26 LEACE Seed-Bootstrap for Stationarity and Change Point

To address the single-run concern for linear concept-erasure results, we rerun LEACE on four canonical encoder-side models (MOMENT, Chronos, PatchTST, GPT4TS) for the two properties with the largest observed drops (stationarity and change point), using the canonical 60/20/20 split, 5 seeds, bootstrap 95% CIs, and a fixed best layer chosen once on seed 0 to avoid layer-reselection leakage across seeds. LEACE itself is fit on the train split only; the post-erasure classifier is refit on the erased training features. Table 37 summarizes the results: all eight (model, property) pairs show large post-erasure accuracy drops (0.461–0.505) with tight CIs. The drop intervals do not overlap zero for any pair, supporting the qualitative claim that stationarity and change-point information lies in a linearly separable subspace recoverable from these representations. *Confirmatory benchmark result.*

Table 37: LEACE seed-bootstrap on the canonical 4-model encoder-side set, at the seed-0 best layer per pair. Reported: pre-erasure and post-erasure accuracy (means over 5 seeds) and the drop with bootstrap 95% CI. Source: `outputs/leace_bootstrap/results.json`. *Confirmatory benchmark result.*

Model	Property	Pre	Post	Drop	95% CI
MOMENT	Stationarity	1.000	0.495	0.505	[0.501, 0.509]
MOMENT	Change point	1.000	0.495	0.505	[0.501, 0.509]
Chronos	Stationarity	1.000	0.526	0.474	[0.465, 0.486]
Chronos	Change point	1.000	0.530	0.470	[0.463, 0.477]
PatchTST	Stationarity	1.000	0.532	0.468	[0.453, 0.480]
PatchTST	Change point	0.999	0.521	0.478	[0.468, 0.492]
GPT4TS	Stationarity	1.000	0.539	0.461	[0.425, 0.496]
GPT4TS	Change point	1.000	0.495	0.505	[0.501, 0.509]

A.27 Linear Intervention Controls: Erasure Sensitivity and Subspace Recovery

To strengthen intervention-based claims beyond “LEACE drop $\neq$ concept dependence,” we perform a three-way validation analysis for each model–property pair at the best probing layer, fitting LEACE exclusively on the training split to prevent data leakage. Critically, we use **LDA directions** (not Ridge SVD) for the recovery test and **Logistic Regression** (not Ridge) for re-classification, avoiding circularity between the probing classifier and the subspace extraction method.

Results in the *linear-subspace sense* are as follows: (1) **Erasure sensitivity**: LEACE erasure causes accuracy to drop (e.g., TimesFM stationarity: $1.00 \rightarrow 0.47$, Chronos change point: $1.00 \rightarrow 0.53$); (2) **Subspace recovery**: projecting onto the LDA concept subspace and re-classifying with Logistic Regression preserves 100% accuracy for 26 of 28 model–property pairs, with two exceptions (Chronos stationarity 81.6%, Chronos change point 69.2%), suggesting that much of the linearly accessible concept information lies in the LDA subspace; (3) **Random control**: erasing a random subspace of equal dimensionality causes zero accuracy drop everywhere. The specificity gap (LEACE drop – random drop) is strictly positive for trend (0.16–0.42), stationarity (0.36–0.53), and change point (0.08–0.51), supporting the interpretation that LEACE targets genuine concept structure. Frequency shows zero specificity across all models, consistent with its distributed, erasure-resistant encoding.

A.28 Real-World Label Stability Analysis

The synthetic-to-real transfer gap (e.g., trend accuracy dropping from 100% to 50% on ETTh1) could reflect either model failure or label unreliability. We measure auto-label stability via bootstrap perturbation across multiple noise levels ($\sigma \in \{0.01, 0.1, 0.5\} \times \text{window std}$, 100 bootstrap trials each). At small perturbations ($\sigma = 0.01$), all labels are stable (> 0.999 agreement). As noise increases to $\sigma = 0.5$, property-dependent stability differences emerge: trend labels become the least stable due to sensitivity of the slope-based labeling threshold to local fluctuations, while stationarity labels remain more robust. Confusion matrix analysis on real-world probes trained and tested *within* ETTh1 (RidgeClassifier, not synthetic-to-real transfer) reveals that MOMENT achieves 84.9% trend accuracy, while GPT4TS reaches only 75.5%, with errors concentrated on weak-slope segments. Note: these within-dataset values differ from the synthetic-to-real transfer results in Table 8 (MOMENT 50.0%), which measure probes trained on synthetic data and evaluated on ETTh1. This analysis reframes the transfer gap as partially a label-definition artifact rather than purely a model deficiency, and motivates concept-specific label quality assessment for real-world evaluation.

A.29 Detailed Limitations

Evaluation Split Protocol and Canonical Rerun. We provide a canonical rerun under train/val/test 60/20/20 with matched sklearn estimators and 5 seeds (Appendix A.3): the central triviality finding is unchanged — four of six properties remain saturated at ≥ 0.999 across all 7 models, and only anomaly shows meaningful variation (0.50–0.75). A retroactive case study on six representative prior-style TSFM probing claims (Appendix A.4) shows that 4/6 become trivial under baseline control, and 1/6 has the baseline *exceeding* the model. An adversarial probing ablation (Appendix A.15) confirms the finding is not a linear-probe artifact: higher-capacity MLP probes on model representations do not exceed the hand-crafted baseline. A generator sensitivity sweep (Appendix A.16) confirms baseline saturation is stable across difficulty parameters.

PCA and Preprocessing. Our PCA sweep (Table 3 and fig. 15) and concept-aware projection analysis (Appendix A.22) confirm that negative seasonality R^2 was an artifact of the linear probe training procedure, not information loss during PCA compression. Full-dimensional Ridge and Ridge on PCA512 features both achieve $R^2 > 0.999$. PCA was fit on all extracted representations (not train-only); since representations are frozen, this is a minor leak, but future runs should fit PCA exclusively on the training split. StandardScaler normalization in baseline and multi-seed scripts was applied before cross-validation splits in the reported results; the released code corrects this via Pipeline. Real-world series are z-score normalized using statistics from the first 80% of the series (temporal train portion) before windowing; windows may overlap (stride 256, length 512), so temporal splits should be used instead of random splits to avoid leakage between adjacent windows.

Linear-Only Interventions. We use linear probes and LEACE for primary analysis. Our nonlinear recovery analysis (Appendix A.25) shows near-zero recovery for frequency after LEACE, but substantial recovery for stationarity and change point and moderate recovery for trend. Accordingly, LEACE-based drops should be interpreted as lower bounds on concept dependence rather than complete erasure of all usable information.

Multi-Seed Validation. Five-seed cross-validated probing with bootstrap 95% CIs shows that stationarity and frequency are effectively deterministic (CI width = 0) across all models, while anomaly is the only property with meaningful variance. Anomaly 95% CIs remain narrow (e.g., MOMENT: 0.561–0.570, Chronos: 0.751–0.755, TimesFM: 0.744–0.767), indicating that the observed anomaly

858 differences are unlikely to be driven only by seed noise. The canonical 5-seed bootstrap re-permutes
859 train/val/test splits but uses a fixed dataset seed; Appendix A.9 resamples both axes (5×5 data-seed
860 $\times$ split-seed cells) and shows that the HC anomaly score is stable at 0.871 [0.862, 0.880], ruling out
861 single-dataset luck.

862 **Sufficient-Statistic Caveat for the Anomaly Inversion.** The canonical anomaly construction (single
863 $\pm 5\sigma$ spike on Gaussian noise) admits a single sufficient statistic: kurtosis (or, equivalently, $\max |x|$)
864 alone reaches 0.859 resp. 0.907 on the same canonical protocol. The HC vector includes kurtosis by
865 design, so the inversion in part says “the protocol’s anomaly task reduces to a single moment that
866 TSFMs do not preserve”. We therefore do not interpret the inversion as evidence that TSFMs fail on *all*
867 anomaly detection tasks; under a harder, mixed-type “realistic” anomaly generator (Appendix A.11)
868 the HC margin shrinks to 0.701. The methodological point is preserved: even on a kurtosis-trivial
869 anomaly task, frozen TSFM representations cannot decode the sufficient statistic, and a stronger
870 ROCKET-style baseline (Appendix A.10) extends the inversion to every hard-variant we tested.

871 **Compute Resources.** All large-scale extraction, probing, intervention, and similarity experiments
872 were run on a single NVIDIA A100 80GB GPU. End-to-end benchmark execution required approxi-
873 mately 48 A100-hours in our internal runs (logged at `outputs/paper/timing.json`), excluding
874 failed exploratory runs and template-only analysis passes; reviewers can verify a single (model,
875 property) cell in under ten minutes via `make smoke`.

876 **Data, Licenses, and Artifact Availability.** All real-world datasets used in TSFMI are public research
877 benchmarks obtained under their original access terms, and all evaluated foundation models are
878 public checkpoints cited in the paper. We do not introduce new human-subject data, private data,
879 or deprecated datasets. Where datasets or checkpoints impose licenses or access restrictions, we
880 follow the original terms rather than redistributing the underlying assets. For anonymous review, any
881 optional code or data artifact is intended to be provided as a separate anonymized supplementary
882 bundle rather than as non-anonymous external links. For camera-ready release, we plan to provide
883 de-anonymized repository links, environment specifications, and usage notes for the benchmark
884 assets.

885 **Scope.** We evaluate seven model families and five real-world datasets. The current benchmark
886 is univariate; extending to multivariate concepts (cross-channel correlation, Granger causality)
887 requires modifying synthetic generators and adding channel-aware probe architectures, though the
888 extraction and evaluation infrastructure is already channel-agnostic through mean-pooling. Extension
889 to medical/financial domains, scaling studies, and mechanistic circuit analysis remains future work.

890 A.30 Benchmark Datasheet

891 Following [44, 45], we provide a structured summary of the TSFMI benchmark:

892 **Motivation.** TSFMI was created to fill the absence of a standardized representation-level evaluation
893 framework for TSFMs — analogous to BERTology [13] for NLP. Existing benchmarks (GIFT-Eval,
894 Monash) evaluate end-task performance only.

895 **Composition.** 11 synthetic datasets (6 standard + 5 hard variants) with exact ground-truth labels, 5
896 real-world datasets (ETTh1, Weather, Electricity, Traffic, Exchange-Rate) with auto-extracted labels,
897 12 model wrappers (7 pre-trained + 5 random baselines), 6 temporal properties.

898 **Collection.** Synthetic data generated via controlled mathematical processes (see
899 `src/datasets/synthetic.py`). Real-world datasets are public benchmarks under original
900 access terms. Stationarity labels use ADF test [46] with fallback to variance-ratio test.

901 **Uses.** Intended for TSFM representation evaluation and comparison. NOT intended for deployment-
902 time behavior modification — intervention tools (LEACE, steering) are diagnostic instruments.

903 **Maintenance.** Community extension via `BaseModelWrapper` interface (4 methods; existing wrap-
904 pers range 121–299 lines, median 173). Versioned releases planned. Anonymized code artifact
905 includes 170 tests.

906 A.31 Extended Error Analysis

907 To better understand where probing fails, we inspect representative error regimes across datasets. In
908 ETTh1, trend confusion is concentrated near weak-slope segments and short local reversals. These
909 segments are difficult even for simple statistical baselines because local windows can be dominated
910 by transient fluctuations rather than global slope. This explains why models that are nearly perfect
911 on synthetic trend can still collapse toward chance-level behavior on a real benchmark when trend
912 strength is low or horizon-dependent.

913 For anomaly probing, false negatives typically occur when outlier amplitude is modest relative to
914 local volatility. False positives occur when deterministic but sharp seasonal transitions are mistaken
915 for anomalies. This pattern is consistent with the low anomaly accuracy range (46–75%) observed
916 even in pre-trained models: linearly accessible representations capture coarse deviations, but do
917 not reliably separate semantically meaningful anomalies from high-curvature normal behavior. In
918 practical applications, this suggests that anomaly pipelines should combine representation probes
919 with residual modeling or temporal context calibration rather than relying on linear readouts alone.

920 Seasonality is a special case. The full-D versus PCA gap indicates that failure under PCA does not
921 imply lack of seasonal information; instead, it indicates that the relevant subspace is not aligned with
922 top-variance axes. In other words, variance ranking is a poor proxy for concept utility in time-series
923 representations. A practical consequence is that dimensionality reduction for deployment should be
924 concept-aware. Techniques such as supervised subspace selection, concept-preserving projection, or
925 hybrid compression (PCA + concept heads) may preserve seasonal information while still reducing
926 storage and latency.

927 A.32 Broader Interpretation and Deployment Guidance

928 The high exploratory CKA between MOMENT and Chronos-Bolt (max-over-layer: 0.997) contrasts
929 with their moderate bootstrap CKA (.35 [.33,.38] at last layer, PCA256), illustrating how layer
930 and reduction choices influence similarity estimates. Under bootstrap CKA, the strongest pairwise
931 alignments are Timer–Moirai (.81) and Chronos–GPT4TS (.80), while most cross-family pairs are
932 near zero. We treat these as hypotheses for follow-up work rather than settled claims.

933 By contrast, GPT4TS appears to trade robustness for flexibility: it can encode useful temporal
934 signals, but these signals are more sensitive to linear interventions. This sensitivity does not make
935 LLM-adapted approaches unusable; rather, it changes operating assumptions. A cautious deployment-
936 oriented workflow would be to (1) evaluate steering sensitivity on held-out data, (2) evaluate concept
937 erasure impact for critical properties, and (3) consider layer selection or limited fine-tuning when
938 sensitivity exceeds a predefined threshold.

939 These diagnostics also suggest a model-governance perspective. For high-stakes tasks, model choice
940 should depend not only on aggregate forecast error but also on representational stability under
941 intervention. Two systems with similar MAE can differ substantially in intervention-sensitive concept
942 usage and failure mode predictability. One possible reporting artifact is a compact interpretability
943 card for TSFMs that includes probing accessibility, LEACE sensitivity, steering sensitivity, and
944 cross-model similarity signatures, while acknowledging that our intervention metrics are currently
945 linear and partly exploratory.

946 Finally, our findings motivate a layered training objective for future TSFMs: optimize predictive
947 performance while regularizing for concept robustness. For example, one could penalize excessive
948 sensitivity to small steering vectors in intermediate layers, or encourage disentanglement between
949 known coupled concepts only where task semantics require separation. Such objectives may improve
950 both interpretability and out-of-distribution reliability, but this remains a hypothesis rather than a
951 validated prescription from the present experiments.

952 A.33 Structural Probe Analysis

953 Table 38 and Figure 11 report Spearman ρ between probe-predicted and true tree depths across layers.
954 MOMENT maintains stable high correlation ($\rho \geq 0.94$) across all layers, while GPT4TS degrades
955 from $\rho = 0.88$ (L0) to $\rho = 0.75$ (L10).

Table 38: Structural probe results (Spearman ρ): Best / Worst Layer.

Model	Best Layer ρ	Worst Layer ρ
MOMENT	0.986 (L17)	0.940 (L0)
Chronos-Bolt	0.923 (L0)	0.855 (L5)
GPT4TS	0.879 (L0)	0.751 (L10)

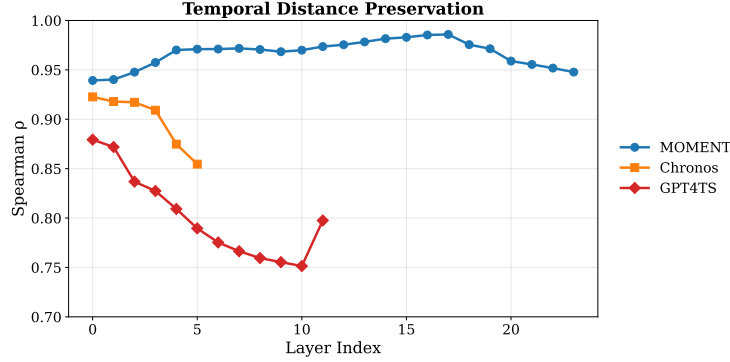


Figure 11: Structural probe correlation across layers. MOMENT remains stable, whereas GPT4TS degrades with depth.

956 A.34 Fine-Tuning Impact on Probing

957 Table 39 and fig. 12 measure how fine-tuning PatchTST-Pre on ETTh1 trend affects probing accessi-
 958 bility. Gains are positive but concentrated in the best middle layer.

959 A.35 Selectivity Analysis

960 Table 40 reports selectivity ($\text{Acc}_{\text{linear}} - \text{Acc}_{\text{MLP}}$) for all models. Near-zero selectivity indicates that
 961 temporal information is trivially linearly accessible, while negative selectivity means the MLP control
 962 probe outperforms the linear probe.

963 A.36 Cross-LEACE Asymmetry Analysis

964 Table 41 quantifies directional dependency between concept pairs by comparing accuracy drops in
 965 both directions. High asymmetry indicates that one concept’s representation depends on the other,
 966 but not vice versa.

967 A.37 Sample Efficiency of Linear Probes

968 We train linear probes on the best layer with varying training set sizes (100–5000 samples). For **trend**
 969 **detection**, Chronos-Bolt, PatchTST-Pre, and GPT4TS achieve 100% accuracy with as few as 100
 970 samples, indicating that trend information is linearly accessible with minimal supervision under this
 971 synthetic setup. MOMENT-PCA512 requires ~ 500 samples to stabilize at 97%, consistent with the

Table 39: Pre-trained vs. fine-tuned PatchTST-Pre trend-classification accuracy (%) on ETTh1, across the three transformer layers. Fine-tuning improves best-layer accuracy by up to +5.7% (layer 2). *Exploratory, single configuration.*

Layer	Pre-trained	Fine-tuned	Improvement
Layer 0	75.5%	79.3%	+3.8%
Layer 1	77.4%	79.3%	+1.9%
Layer 2	77.4%	83.0%	+5.7%

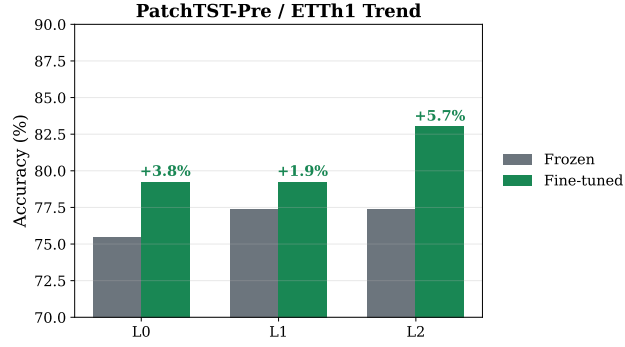


Figure 12: Trend detection accuracy after fine-tuning PatchTST-Pre on ETTh1. Gains are positive but concentrated in the best middle layer.

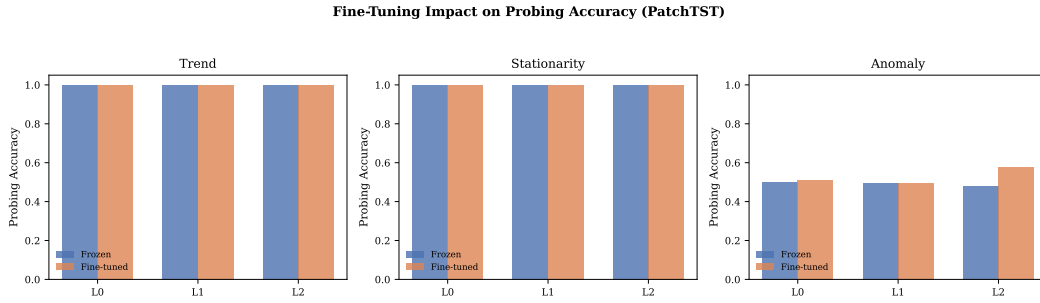


Figure 13: Frozen-vs-fine-tuned probe accuracy. Fine-tuning gains concentrate on anomaly, while trend and stationarity remain near ceiling.

Table 40: Selectivity (Linear Accuracy – MLP Control Accuracy) for all models.

Model	Trend	Season. (R^2)	Frequency	Station.	Anomaly	Change Pt.
MOMENT-PCA512	+0.000	-2.117	+0.000	+0.000	+0.002	+0.000
Chronos-Bolt	+0.000	-0.000	+0.000	+0.000	-0.010	+0.000
PatchTST-Pre	+0.000	-0.000	+0.000	+0.000	+0.010	+0.000
GPT4TS-PCA512	+0.000	-2.040	+0.000	+0.000	-0.006	+0.000
Timer-Base	+0.000	—	+0.000	+0.000	+0.000	+0.000
TimesFM-500M	+0.000	-0.035	+0.000	+0.000	-0.110	+0.000
Moirai-Small	+0.000	-0.002	+0.000	+0.000	-0.090	+0.000
PatchTST-Rnd	+0.000	-0.001	+0.000	-0.536	-0.102	-0.002
iTransf.-Rnd	+0.000	-0.001	+0.000	-0.002	-0.020	+0.000

Table 41: Cross-LEACE Asymmetry Analysis. Absolute difference in accuracy drop when erasing concept A to evaluate B vs. erasing B to evaluate A.

Model	Concept Pair	Forward Drop	Reverse Drop	Asymmetry
MOMENT	Change Pt. $\leftrightarrow$ Stationarity	40.9%	40.7%	0.2%
MOMENT	Change Pt. $\leftrightarrow$ Trend	36.0%	0.0%	36.0%
MOMENT	Stationarity $\leftrightarrow$ Trend	3.3%	12.2%	8.9%
GPT4TS	Change Pt. $\leftrightarrow$ Stationarity	26.2%	47.7%	21.5%
GPT4TS	Change Pt. $\leftrightarrow$ Trend	34.0%	38.0%	4.0%
GPT4TS	Stationarity $\leftrightarrow$ Trend	0.0%	17.6%	17.6%

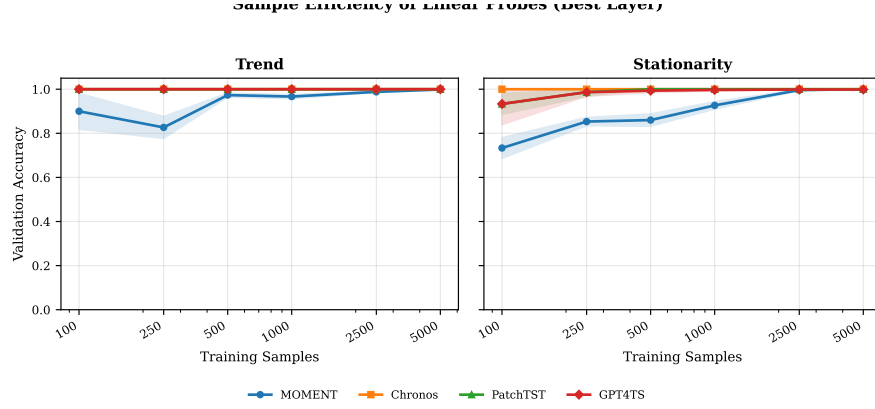


Figure 14: Sample efficiency of linear probes at the best layer. Trend and stationarity saturate with surprisingly little supervision in most families.

Table 42: Real-world LEACE erasure: best-layer accuracy before/after erasure and accuracy drop.

Model	Property	ETTh1				Weather			
		Before	After	Drop	Layer	Before	After	Drop	Layer
MOMENT	Trend	96.2	92.5	3.8	L10	99.0	81.0	18.0	L6
MOMENT	Station.	96.2	75.5	20.8	L12	88.5	72.0	16.5	L6
Chronos	Trend	96.2	86.8	9.4	L0	99.5	51.5	48.0	L0
Chronos	Station.	100.0	98.1	1.9	L0	98.5	95.0	3.5	L5
PatchTST	Trend	98.1	90.6	7.5	L1	100.0	59.5	40.5	L2
PatchTST	Station.	100.0	94.3	5.7	L2	93.5	93.0	0.5	L2
GPT4TS	Trend	98.1	83.0	15.1	L11	98.5	55.6	42.9	L6
GPT4TS	Station.	94.3	79.2	15.1	L2	90.3	86.9	3.3	L1

PCA-reduced probe setup requiring more data to converge. For **stationarity**, Chronos-Bolt achieves 100% from 100 samples, while PatchTST-Pre and GPT4TS start at 93% and converge by 500 samples. MOMENT-PCA512 shows the slowest convergence, requiring ~ 2500 samples to reach near-perfect accuracy. This gap reinforces the PCA bottleneck finding: dimensionality reduction forces the linear probe to compensate with more training data.

A.38 Real-World LEACE Erasure

Table 42 shows LEACE concept erasure applied to real-world datasets (ETTh1, Weather) for trend and stationarity. Unlike synthetic data where trend drops are often modest, real-world **Weather trend** shows large degradations: Chronos-Bolt loses 48.0% accuracy, GPT4TS 42.9%, and PatchTST-Pre 40.5%, while MOMENT shows an 18.0% drop. This indicates that real-world trend representations can also be sensitive to linear erasure of trend subspaces. On ETTh1, trend erasure effects are smaller: GPT4TS drops 15.1%, Chronos 9.4%, PatchTST 7.5%, and MOMENT only 3.8%.

Stationarity erasure shows varied sensitivity: MOMENT drops 20.8% on ETTh1 and GPT4TS 15.1%, while Chronos-Bolt is nearly unaffected (1.9%). On Weather, stationarity erasure has minimal impact for Chronos, PatchTST, and GPT4TS ($\leq 3.5\%$) but remains substantial for MOMENT (16.5%), indicating that stationarity is encoded more diffusely in some families than in others.

A.39 PCA Dimension Sweep

Figure 15 shows the complete PCA dimension sweep for seasonality regression. Across all tested dimensions (64–512), PCA-reduced representations yield negative R^2 , suggesting that top-variance principal components are poorly aligned with the seasonality-relevant subspace in this probe setup. The monotonically worsening R^2 with more components further suggests that additional PCA dimensions introduce variance-capturing directions that interfere with seasonality prediction.

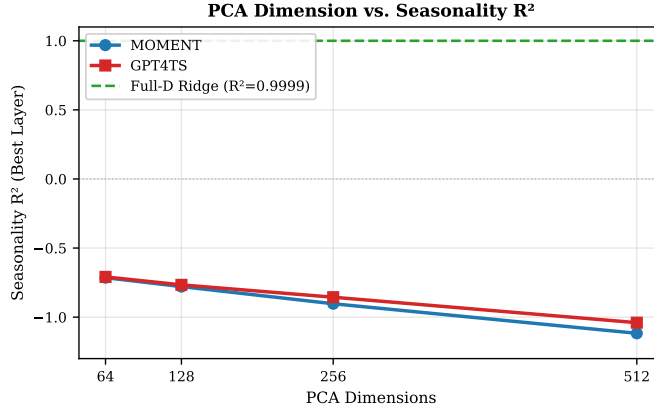


Figure 15: PCA dimension sweep for seasonality R^2 . All PCA dimensions remain negative, whereas full-dimensional Ridge recovers near-perfect performance.

A.40 t-SNE Representation Visualizations

Figures 16 and 17 show t-SNE projections of Layer 0 and best layer representations for trend and stationarity classification. Chronos-Bolt and PatchTST-Pre show clear class separation from Layer 0, consistent with their high probing accuracy. GPT4TS shows less distinct boundaries, reflecting its PCA-compressed representation. MOMENT-PCA512 shows moderate separation that improves in later layers for trend.

A.41 UMAP Representation Visualizations

Figures 18 and 19 complement the t-SNE projections with UMAP, which better preserves global structure. The results are qualitatively consistent: Chronos-Bolt and PatchTST-Pre show clear class separation from Layer 0, while GPT4TS-PCA512 shows diffuse boundaries.

A.42 PatchTST Attention Pattern Analysis

Figure 20 shows attention entropy and mean attention distance across PatchTST-Pre’s 3 layers for trend and seasonality. For **trend**, attention entropy dips in Layer 1 (3.43 vs. 3.67 in L0), suggesting the middle layer specializes with sharper attention patterns. Per-class analysis reveals the *Up* class consistently receives higher entropy (more diffuse attention) than *Down*, potentially reflecting asymmetric pattern complexity. Mean attention distance also decreases in Layer 1, indicating more localized processing. For **seasonality**, entropy and distance follow similar layer-wise patterns but without per-class differentiation (binary property).

A.43 Anomaly Noise Robustness

Figure 21 shows probe accuracy as anomaly magnitude (signal-to-noise ratio) varies from 1.0 to 8.0. **Chronos-Bolt** shows the largest observed robustness gain, scaling from 48.6% at magnitude 1.0 (near-random) to 88.6% at magnitude 8.0. MOMENT follows at 70.8%, while PatchTST-Pre (54.4%) and GPT4TS (54.3%) remain near random even at high SNR. In this setup, decoder-style models appear to encode anomaly signal strength more accessibly than patch-based models.

A.44 Downstream Reconstruction Impact of Concept Erasure

Figure 22 quantifies how LEACE erasure affects MOMENT’s downstream reconstruction quality. For **trend** erasure, early layers show minimal impact (L0: +1.0% MSE), middle layers show the most degradation (L11: +5.6%), and the final layer shows an unexpected MSE *decrease* (−17.2%). This suggests that trend information in late layers may be redundant or interfere with reconstruction, and erasing it may act as a form of regularization. For **stationarity**, the pattern is similar: early erasure

TSNE Projections: Synthetic Trend (Layer 0 vs Best Layer)

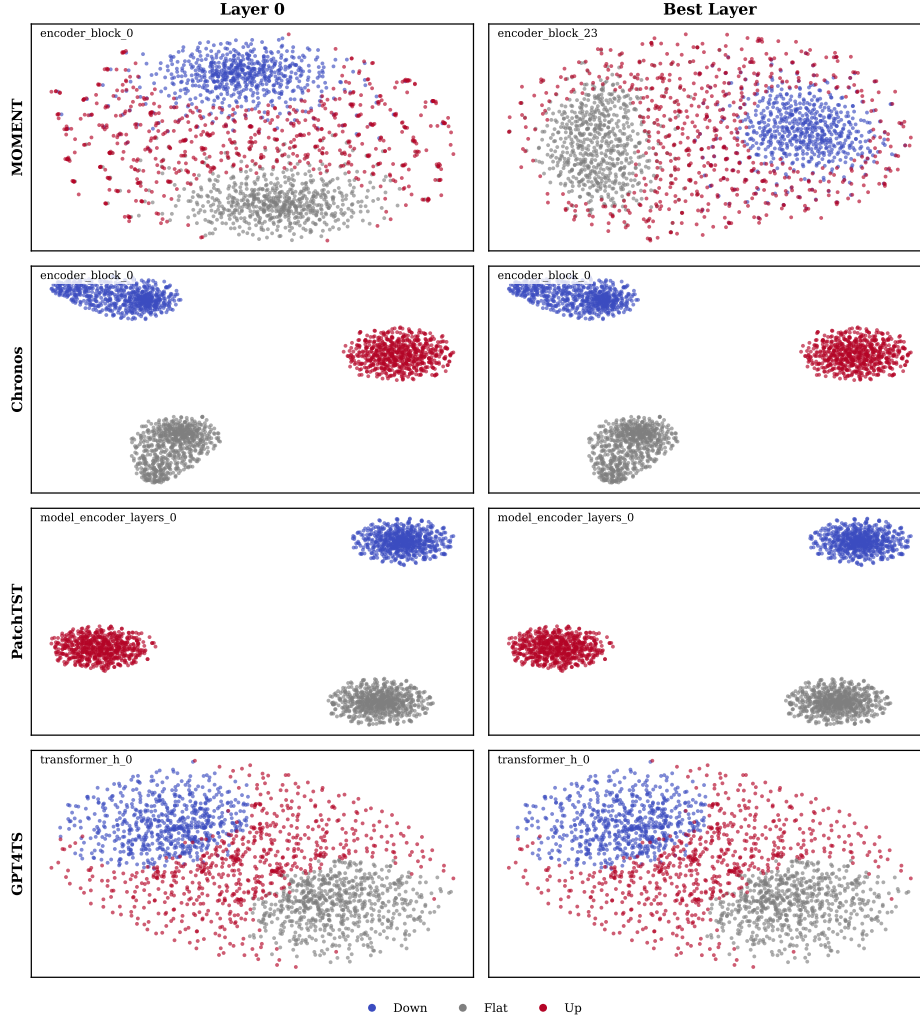


Figure 16: t-SNE projections for synthetic trend at Layer 0 versus the best layer. Separation is already visible early for the clearest-performing families.

has modest impact (L0: +5.8%), while middle-layer erasure causes the largest degradation (L11: +18.0%). This places the largest observed stationarity sensitivity in middle encoder blocks.

A.45 Downstream Forecasting Impact (Chronos-Bolt)

Figure 23 extends the interventional analysis to Chronos-Bolt’s forecasting task (64-step horizon). **Trend** erasure causes very large degradation, especially at early layers: L0 produces +7808% MSE increase, L2 +644%, and L5 +1571%. This is much larger than MOMENT’s trend erasure effects, indicating that Chronos-Bolt’s forecasting relies heavily on early-layer trend encoding. **Stationarity** erasure has negligible impact (−1.0% at L0, +8.7% at L2, −2.1% at L5), suggesting stationarity information is either redundant for forecasting or encoded non-linearly.

A.46 Downstream Output Impact (GPT4TS)

Figure 24 measures the output divergence of GPT4TS (frozen GPT-2) after LEACE erasure, complementing the reconstruction (MOMENT) and forecasting (Chronos-Bolt) analyses. Unlike native TSFMs, GPT4TS exhibits a distinct pattern: **stationarity** erasure causes the largest output disruption

TSNE Projections: Synthetic Stationarity (Layer 0 vs Best Layer)

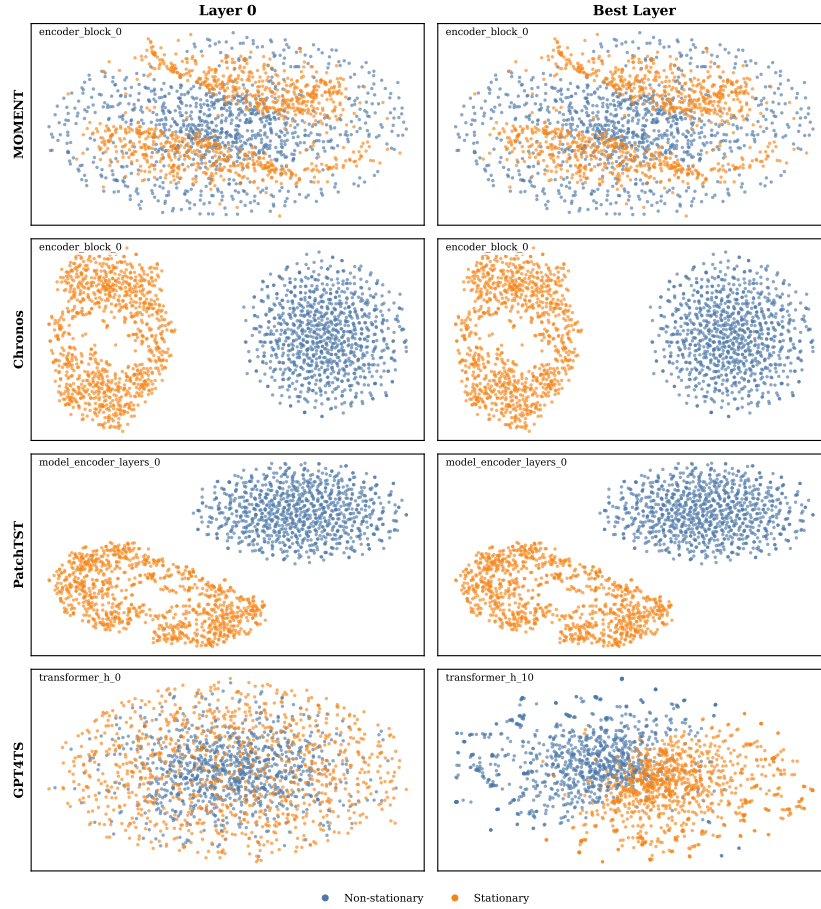


Figure 17: t-SNE projections for synthetic stationarity at Layer 0 versus the best layer. Class separation sharpens only modestly with depth.

(55.1% divergence at L5), substantially exceeding trend erasure (37.5% at L5). This asymmetry is reversed compared to Chronos-Bolt, where trend erasure dominates. Notably, late-layer (L11) erasure has the least impact for both properties (4.2% trend, 24.4% stationarity), suggesting the frozen LLM backbone consolidates temporal concepts in middle layers. This mid-layer concentration mirrors MOMENT’s reconstruction pattern but contrasts with Chronos-Bolt’s early-layer reliance, revealing architecture-specific intervention pathways.

A.47 Downstream Forecasting Impact (TimesFM)

Figure 25 extends interventional analysis to TimesFM, the largest decoder-only model (494M, 50 layers). **Trend** erasure produces the most extreme relative effects in our study: even early-layer erasure (L0) increases forecast MSE by +195,775%, escalating to +413,099% at L49. Because these are relative MSE changes, they should be interpreted as instability indicators rather than directly comparable effect sizes across models. **Stationarity** erasure, by contrast, has negligible impact (−1.6% to +5.0%), mirroring Chronos-Bolt’s pattern and suggesting that, in this setup, decoder-only forecasting is much more sensitive to trend-subspace removal than to stationarity-subspace removal.

A.48 Downstream Output Impact (Moirai v2)

Figure 26 measures Moirai v2’s output divergence after LEACE erasure. Unlike decoder-only models, Moirai shows a **reversed priority**: **stationarity** erasure causes larger output disruption (peak 10.9%

UMAP Projections: Synthetic Trend (Layer 0 vs Best Layer)

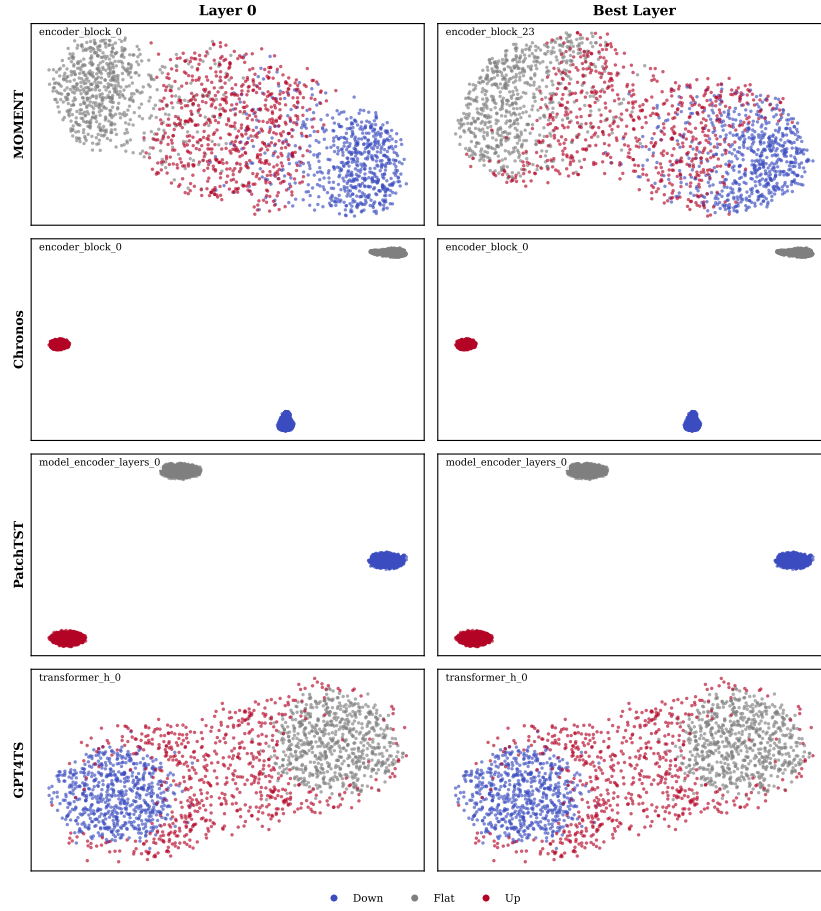


Figure 18: UMAP projections for synthetic trend at Layer 0 versus the best layer. Global structure mirrors the t-SNE trends.

Table 43: Timer (84M) probing results: best-layer accuracy/ R^2 for each property.

Model	Trend	Season. (R^2)	Frequency	Station.	Anomaly	Change Pt.
Timer-Base	1.000	0.998	1.000	1.000	0.460	1.000

divergence at L1) than **trend** erasure (peak 3.2% at L0). Both properties show a decreasing impact with depth (L0 most affected, L5 least), consistent with Moirai’s compact encoder architecture concentrating intervention-sensitive information in early layers. The overall magnitudes are modest compared to TimesFM’s large trend sensitivity, reflecting Moirai’s smaller parameter count (11.4M vs. 494M) and shallower error amplification chain.

A.49 Timer Model Probing Results

Timer [47], a decoder-only generative TSFM with 84M parameters and 8 layers, shows probing performance consistent with other pre-trained models. **Trend**, **frequency**, **stationarity**, and **change point** are perfectly linearly separable from every layer (100% accuracy). **Seasonality** regression reaches $R^2 = 0.998$ at Layer 5. However, **anomaly detection** remains weak (best 47.7% averaged over 3 seeds at Layer 5), among the lowest of pre-trained models, indicating that point anomalies are challenging across architectural families.

UMAP Projections: Synthetic Stationarity (Layer 0 vs Best Layer)

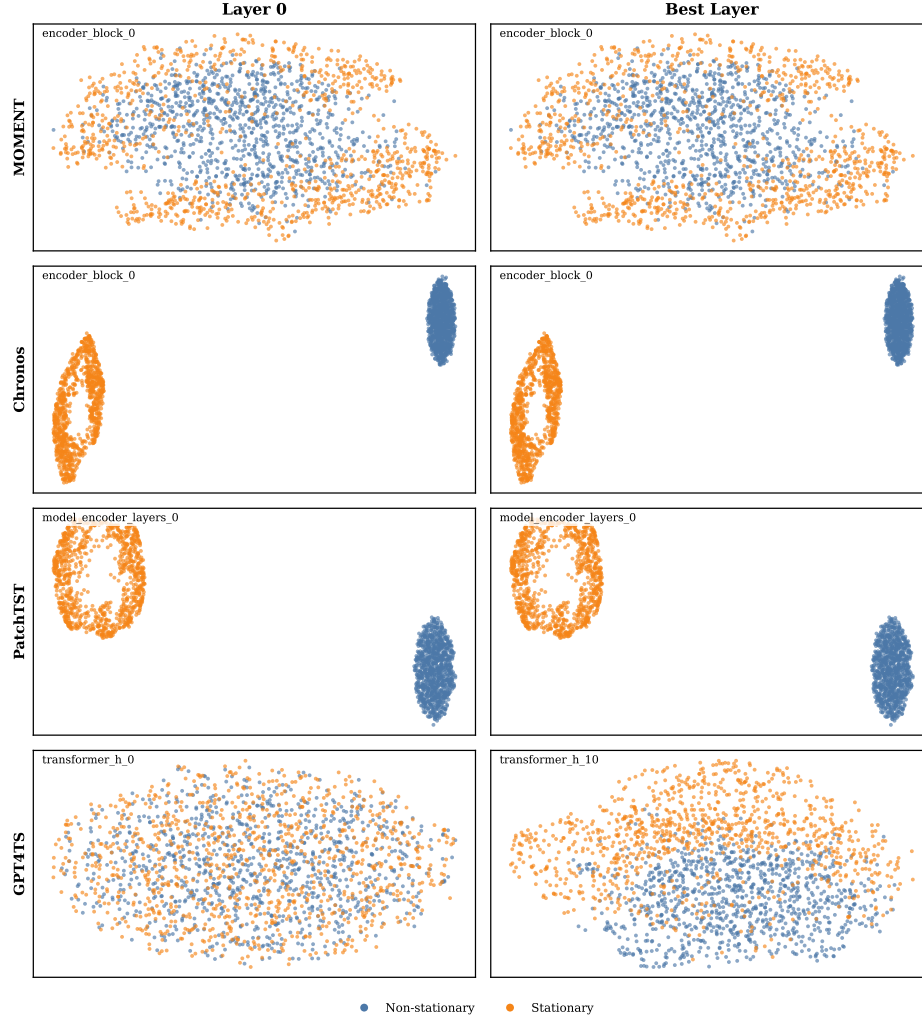


Figure 19: UMAP projections for synthetic stationarity at Layer 0 versus the best layer. Global structure remains more diffuse than for trend.

1066 A.49.1 Timer LEACE Erasure

1067 LEACE erasure on Timer-Base (mean-pooled representations, $d=1024$) reveals decoder-specific
1068 erasure patterns:

1069 Timer concentrates trend, frequency, and stationarity erasure at **L0**, while change-point vulnerability
1070 peaks at **L7**. This early-layer concentration mirrors Chronos-Bolt’s pattern, consistent with decoder-
1071 only architectures encoding structural properties immediately. Seasonality is most vulnerable at L5
1072 (R^2 drops to -0.193), while anomaly remains resistant to erasure (4% drop), suggesting it is not
1073 linearly encoded in any single direction.

1074 A.50 TimesFM and Moirai Probing Results

1075 TimesFM (494M, 50 decoder layers) and Moirai (11.4M, 6 encoder layers) extend our analysis to the
1076 largest and smallest models in the study.

1077 TimesFM concentrates stationarity and change-point erasure at **L0**, mirroring Timer’s early-layer
1078 pattern. With 50 layers, TimesFM shows the deepest model analyzed, yet its LEACE profile broadly

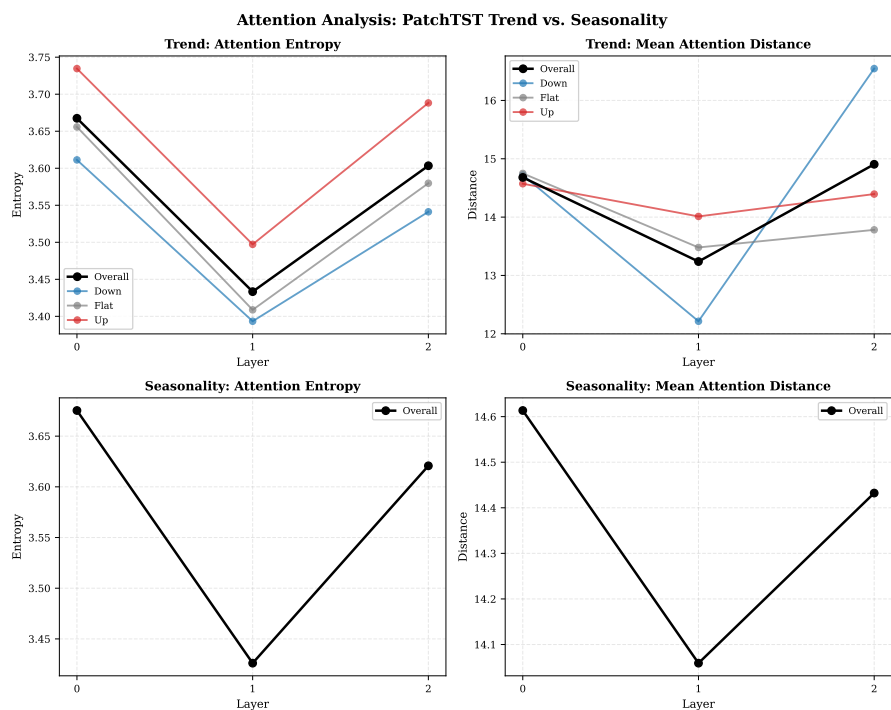


Figure 20: PatchTST attention entropy and mean distance across layers. The middle layer shows the sharpest, most localized attention patterns.

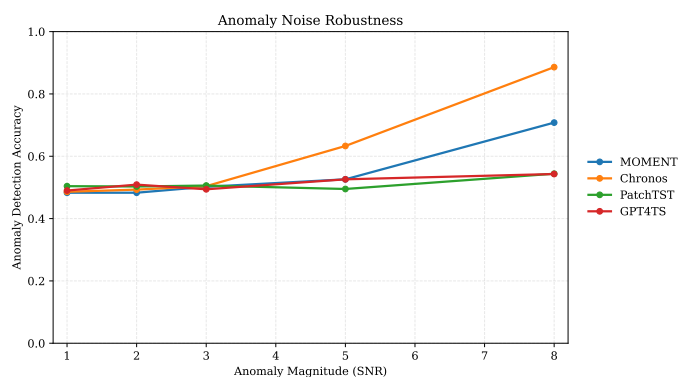


Figure 21: Anomaly detection accuracy versus anomaly magnitude (SNR) across model families. Decoder-style models benefit most from stronger anomaly signal.

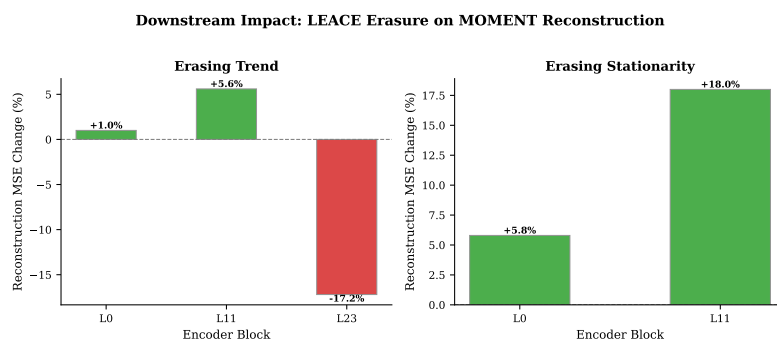


Figure 22: Reconstruction MSE change (%) after LEACE erasure at MOMENT layers. Middle-layer erasure produces the largest downstream degradation in this analysis.

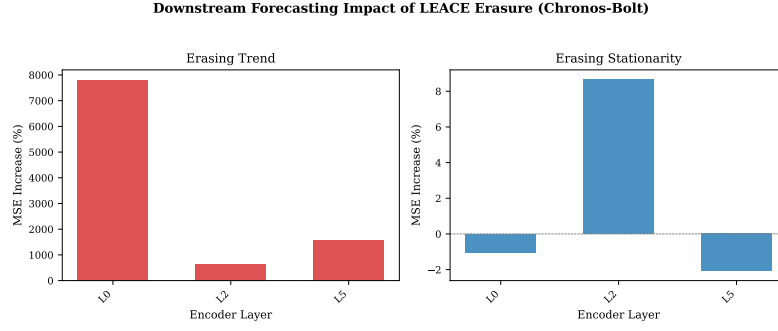


Figure 23: Forecasting MSE change (%) after LEACE erasure at Chronos-Bolt layers. Early trend erasure has the largest downstream effect.

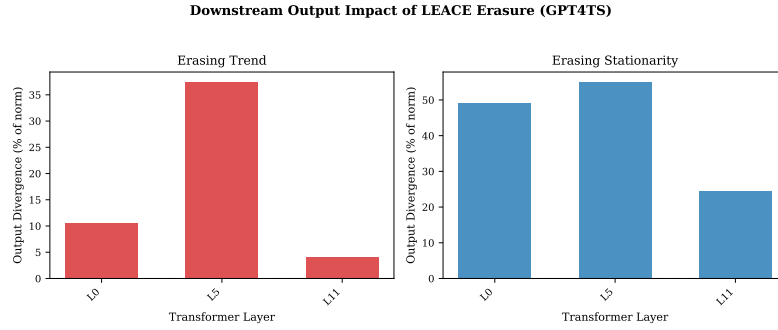


Figure 24: Output divergence (% of output norm) after LEACE erasure at GPT4TS layers. Stationarity removal perturbs outputs more than trend removal.

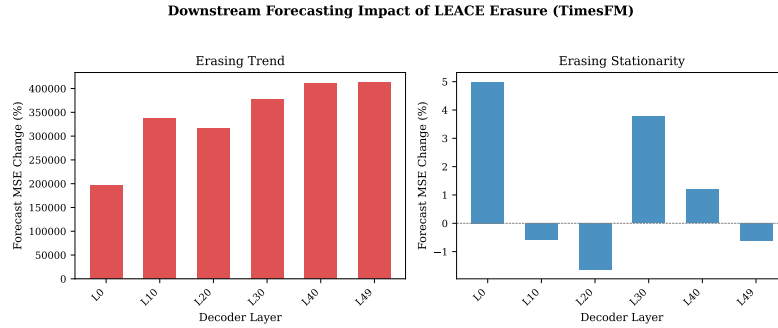


Figure 25: Forecasting MSE change (%) after LEACE erasure at TimesFM decoder layers. Trend erasure causes very large degradation at all layers.

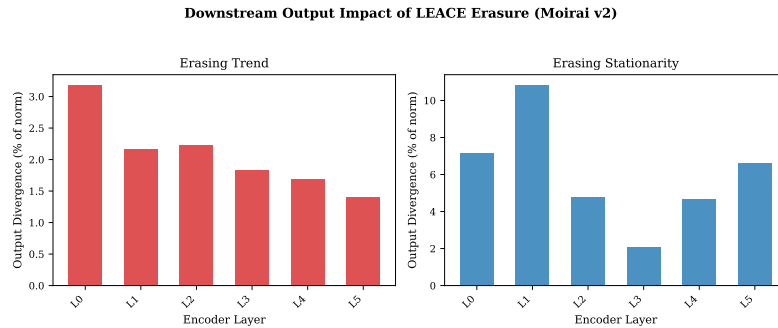


Figure 26: Output divergence (% of output norm) after LEACE erasure at Moirai layers. Stationarity has the stronger downstream footprint.

Table 44: Timer-Base LEACE erasure: accuracy/ R^2 before and after erasure at the max-drop layer.

Property	Before	After	Drop	Layer
Trend	1.000	0.665	0.335	L0
Seasonality (R^2)	0.800	-0.193	0.993	L5
Frequency	1.000	0.800	0.200	L0
Stationarity	0.995	0.365	0.630	L0
Anomaly	0.680	0.640	0.040	L3
Change Point	1.000	0.365	0.635	L7

Table 45: TimesFM LEACE erasure: accuracy/ R^2 before and after erasure at the max-drop layer (reported every 5th layer).

Property	Before	After	Drop	Layer
Trend	1.000	0.630	0.370	L5
Seasonality (R^2)	0.956	-0.029	0.985	L25
Frequency	1.000	0.635	0.365	L0
Stationarity	1.000	0.300	0.700	L0
Anomaly	0.800	0.645	0.155	L25
Change Point	1.000	0.470	0.530	L0

resembles the other decoder-only models. Moirai, despite being the smallest model (11.4M), shows the largest LEACE sensitivity to stationarity (70% drop) and change points (65% drop), suggesting compact encoders may rely more heavily on single linear directions for concept encoding.

A.51 Downstream Forecasting Impact (TimesFM)

Table 47 provides the full per-layer downstream impact data for TimesFM. Trend erasure MSE increase grows monotonically from L0 (+195,775%) to L49 (+413,099%), while stationarity remains within $\pm 5\%$ at all layers. We interpret this as an extreme sensitivity pattern under relative MSE reporting, not as a calibrated estimate of causal effect magnitude.

A.52 Downstream Output Impact (Moirai v2)

Table 48 shows Moirai’s per-layer output divergence. Stationarity erasure peaks at L1 (10.9%) then declines, while trend erasure decreases monotonically from L0 (3.2%) to L5 (1.4%).

Table 46: Moirai LEACE erasure: accuracy/ R^2 before and after erasure (max-drop layer).

Property	Before	After	Drop	Layer
Trend	1.000	0.630	0.370	L0
Seasonality (R^2)	0.996	-0.020	1.016	L5
Frequency	1.000	0.505	0.495	L0
Stationarity	1.000	0.300	0.700	L0
Anomaly	0.745	0.495	0.250	L0
Change Point	1.000	0.350	0.650	L0

Table 47: TimesFM downstream forecasting impact: MSE change (%) after LEACE trend or stationarity erasure.

Layer	Trend MSE Δ (%)	Stationarity MSE Δ (%)
L0	+195,775	+5.0
L10	+336,412	−0.6
L20	+316,872	−1.6
L30	+377,625	+3.8
L40	+410,999	+1.2
L49	+413,099	−0.6

Table 48: Moirai v2 downstream output divergence (% of output norm) after LEACE erasure.

Layer	Trend (%)	Stationarity (%)
L0	3.2	7.2
L1	2.2	10.9
L2	2.2	4.8
L3	1.8	2.1
L4	1.7	4.7
L5	1.4	6.6

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state the headline empirical claim (hand-crafted features outperform every evaluated TSFM on anomaly probing, while four other properties saturate non-model baselines under the canonical protocol), and the methodological claim (representation-level claims about TSFMs require explicit non-model controls). Both are directly supported by the canonical results table in the main body and by the appendix manifest, retroactive case study, adversarial probing, and seed-bootstrap LEACE artifacts.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: The paper discusses limitations including PCA-compressed views for some models, protocol-level rather than raw-state cross-family comparisons, and weak synthetic-to-real transfer; Appendix A.1 also states scope limitations explicitly.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: This submission is an empirical benchmark paper and does not claim new theorems or proofs.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: Primary confirmatory results use a canonical 60/20/20 train/val/test protocol with best-layer selection on val, final reporting on held-out test, matched sklearn estimators,

1120 5 seeds, and bootstrap 95% CIs. Known preprocessing caveats (PCA fit scope, extraction
1121 heterogeneity across model families) are disclosed in the limitations and appendix. Legacy
1122 PyTorch probing results (80/10/10 split) are retained only as auxiliary diagnostics.

1123 5. Open access to data and code

1124 Question: Does the paper provide open access to the data and code, with sufficient instruc-
1125 tions to faithfully reproduce the main experimental results, as described in supplemental
1126 material?

1127 Answer: [Yes]

1128 Justification: An anonymized code artifact is provided as supplementary material, containing
1129 all source code, test suite (170 tests), synthetic data generators, extraction pipelines, and
1130 evaluation scripts with reproduction instructions.

1131 6. Experimental setting/details

1132 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
1133 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1134 Answer: [Yes]

1135 Justification: The paper reports the canonical confirmatory setup (60/20/20 train/val/test,
1136 matched sklearn LogisticRegression/Ridge estimators, 5 seeds, bootstrap 95% CIs), the
1137 temporal properties and datasets, and the intervention settings. Architecture details and
1138 supplementary settings are in the appendix. Baseline controls and model probes share the
1139 identical scoring pipeline in the canonical protocol; legacy PyTorch probing at 80/10/10 is
1140 retained only for auxiliary diagnostics and is disclosed as such.

1141 7. Experiment statistical significance

1142 Question: Does the paper report error bars suitably and correctly defined or other appropriate
1143 information about the statistical significance of the experiments?

1144 Answer: [Yes]

1145 Justification: Primary confirmatory results (canonical baseline-control table and per-model
1146 canonical rerun) report 5-seed bootstrap 95% confidence intervals on held-out test scores;
1147 seed-bootstrap LEACE results for the four canonical encoder-side models are also reported
1148 in the appendix. The legacy 3-seed PyTorch probe table is retained only as auxiliary
1149 diagnostic and is labeled as such. Cross-property entanglement, downstream impact, and
1150 other single-run results are explicitly labeled exploratory and are not used as primary
1151 evidence. Real-world comparisons with small n are reported as reference diagnostics rather
1152 than rankings.

1153 8. Experiments compute resources

1154 Question: For each experiment, does the paper provide sufficient information on the com-
1155 puter resources (type of compute workers, memory, time of execution) needed to reproduce
1156 the experiments?

1157 Answer: [Yes]

1158 Justification: The appendix states that the benchmark was run on a single NVIDIA A100
1159 80GB GPU and reports the approximate end-to-end compute budget for extraction, probing,
1160 intervention, and similarity analysis.

1161 9. Code of ethics

1162 Question: Does the research conducted in the paper conform, in every respect, with the
1163 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

1164 Answer: [Yes]

1165 Justification: The work analyzes public models and public benchmark datasets only, does
1166 not involve private or human-subject data, and explicitly frames intervention methods as
1167 diagnostic tools rather than undisclosed deployment-time manipulation mechanisms.

1168 10. Broader impacts

1169 Question: Does the paper discuss both potential positive societal impacts and negative
1170 societal impacts of the work performed?

1171 Answer: [\[Yes\]](#)

1172 Justification: The paper discusses interpretability benefits for safer model diagnosis and
 1173 selection, while also noting that steering/erasure techniques could be misused if applied
 1174 outside controlled evaluation settings.

1175 **11. Safeguards**

1176 Question: Does the paper describe safeguards that have been put in place for responsible
 1177 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
 1178 image generators, or scraped datasets)?

1179 Answer: [\[N/A\]](#)

1180 Justification: The paper does not release new generative models or scraped datasets; it
 1181 evaluates already public time-series models and public benchmark data.

1182 **12. Licenses for existing assets**

1183 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
 1184 the paper, properly credited and are the license and terms of use explicitly mentioned and
 1185 properly respected?

1186 Answer: [\[Yes\]](#)

1187 Justification: The paper cites the original model and dataset papers and now states in the
 1188 appendix that the benchmark uses only public datasets and checkpoints under their original
 1189 access terms, with no deprecated or private datasets introduced by this submission.

1190 **13. New assets**

1191 Question: Are new assets introduced in the paper well documented and is the documentation
 1192 provided alongside the assets?

1193 Answer: [\[Yes\]](#)

1194 Justification: An anonymized code artifact is provided as supplementary material with a
 1195 README containing installation and reproduction instructions, 170 automated tests, and a
 1196 Benchmark Datasheet (Appendix A.30) following the Datasheets for Datasets framework.
 1197 The benchmark includes synthetic data generators, model wrappers, and evaluation scripts.

1198 **14. Crowdsourcing and research with human subjects**

1199 Question: For crowdsourcing experiments and research with human subjects, does the paper
 1200 include the full text of instructions given to participants and screenshots, if applicable, as
 1201 well as details about compensation (if any)?

1202 Answer: [\[N/A\]](#)

1203 Justification: The work does not involve crowdsourcing or human subjects.

1204 **15. Institutional review board (IRB) approvals or equivalent for research with human
 1205 subjects**

1206 Question: Does the paper describe potential risks incurred by study participants, whether
 1207 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
 1208 approvals (or an equivalent approval/review based on the requirements of your country or
 1209 institution) were obtained?

1210 Answer: [\[N/A\]](#)

1211 Justification: The work does not involve human subjects.

1212 **16. Declaration of LLM usage**

1213 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 1214 non-standard component of the core methods in this research? Note that if the LLM is used
 1215 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
 1216 scientific rigor, or originality of the research, declaration is not required.

1217 Answer: [\[N/A\]](#)

1218 Justification: LLMs are not part of the benchmark methodology, model training, probing,
 1219 intervention, or analysis pipeline described in the paper.